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Scale And Conformal Invariance in 2D Quantum Field Theories

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Class:

2nd year , 2nd semester
Master of Science (M.Sc.) Course

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Academic Year: 2023/2025

CERTIFICATE

Certified that this project titled **“Scale and Conformal Invariance in 2D Quantum Field Theory”** is the bonafied work of **Shubham Das**, a student of Theoretical Physics, The Institute of Mathematical Sciences, Chennai, who carried out the project under my supervision in partial fulfillment of requirements for the MSc degree under the Homi Bhaba National Institute, Mumbai.

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Acknowledgment

I would like to express my deepest gratitude to my advisor, Professor Sujay K. Ashok, for his invaluable guidance, unwavering support, and expertise throughout the entire process of learning and writing this thesis. His mentorship has been instrumental in shaping my understanding of conformal field theories and navigating the complexities of this fascinating field.

Abstract

Scale and conformal transformations are mathematically distinct transformations, so we have no reason to believe that scale invariance implies conformal invariance. Still, many quantum field theories that we know are scale invariant also show conformal invariance. We will demonstrate the relation between scale and conformal invariance in two dimensions. We will show that under some assumptions, the scale invariant quantum field theory in two dimensions has enhanced conformal symmetry. The proof relies on the rich structure of the renormalization group flows in two dimensions. Then we will show how the scale and conformal invariance constraints the trace of the stress tensor and how this provides an insight into when scale invariance implies conformal invariance. Finally, we will see the nonlinear sigma model in two dimensions with non-compact or quasi-Ricci flat target space serves as a counterexample to the claim that scale invariance implies conformal invariance in two dimensions.

Contents

List of Figures	vii
1 Introduction	1
2 Scale And Conformal Invariance In Quantum Field Theory	3
2.1 Introduction	3
2.2 The Conformal Group	3
2.3 Representations of the Conformal Group	4
2.4 Symmetries and Energy-Momentum Tensor	5
2.5 Unitarity	6
2.6 Consequence of Scale and Conformal Invariance	8
3 C-theorem	11
3.1 Introduction	11
3.2 The Renormalization Group in Two-Dimensional Field Theory	12
3.2.1 Homogeneous Renormalization Group	13
3.2.2 Local Renormalization Group	16
3.3 Perturbation Theory About Fixed Point	17
4 Does Scale Invariance imply Conformal Invariance?	29
4.1 Introduction	29
4.2 The Construction of a Traceless Energy-Momentum Tensor	29
4.3 Scale vs. Conformal Invariance: Examples	33
4.4 Zamolodchikov's c-theorem	35
5 Non-Linear Sigma Model	37
5.1 Introduction	37
5.2 Conformally Invariant Field Theory in Two Dimensions	37
5.2.1 The Nambu-Goto and Polyakov Action	37
5.2.2 Weyl and Conformal Invariance	39
5.2.3 The Quantum Anomaly	41
5.3 Background Field Expansion and Riemann Normal Coordinates	43
5.4 One Loop Calculation of Weyl Anomaly	46
6 Conclusion	51
Bibliography	53

List of Figures

1.1	Scale and conformal transformation in two dimensions.	1
2.1	Map of cylinder onto a plane	7
5.1	One-loop diagram contributing to trace. Here, the wavy lines represent the classical fields, the normal lines are the propagators, and the "x" mark represents the T_{++} insertion.	47

Chapter 1

Introduction

Symmetries play a profound role in the study of systems. They govern everything from fundamental forces to conservation laws. They are transformations that leave the physical system unchanged, revealing deep insights about the seemingly unrelated phenomena. Emmy Noether showed that corresponding to every continuous symmetry, there is a conserved quantity. This allows us to determine conserved quantities just by observing the symmetry laws, which drastically simplifies the analysis. The work of Lie and Cartan helped extend this notion of symmetries to applications in quantum mechanics. Wigner demonstrated how these group-theoretic concepts can be used to represent the elementary particles as representations of the Lorentz and Poincaré group. Einstein's special and general theories of relativity are motivated by symmetry considerations. We know that the quantum electrodynamics (QED) action has gauge invariance, but if we treat gauge symmetry as fundamental, we can derive all results of QED from this principle. This gauge symmetry led to the construction of other interesting Lagrangians. These field theories were first constructed by Yang and Mills, generalizing QED profoundly.

We are familiar with continuous symmetries like translations and rotations, but there are two more symmetries that a system can possess: invariance under scale transformation and conformal transformation.

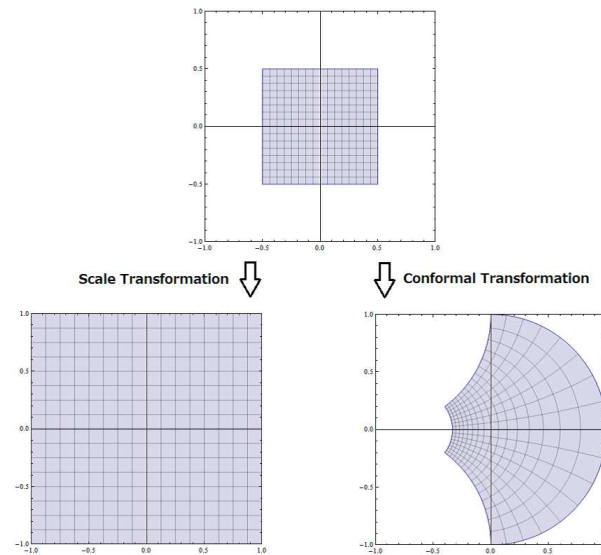


FIGURE 1.1: Scale and conformal transformation in two dimensions.

A scale transformation increases or decreases the size of the system by a scale factor that is the same in all directions. Conformal transformation leaves the angle between the two curves the same. As we will see, scale invariance does not imply conformal invariance, at least at the level of mathematical definitions. We will demonstrate the mathematical distinction between the group structure with scale transformation and conformal transformation. We will try to

understand the relation between these two transformations. We have no reason to believe that scale invariance implies conformal invariance, but the majority of the quantum field theories that we are familiar with show conformal invariance. We will try to understand why this enhancement to conformal invariance from scale invariance takes place in two dimensions.

The idea of scale invariance in physics stems from the renormalization group flow. Wilson's renormalization group is a successive application of integrating out fields and scale transformation. We can describe the long-range physics by integrating out short degrees of freedom, dealing with an effective field theory. We will use this idea of renormalization group flows to show the existence of c -function which continuously decreases along the renormalization group trajectory. This function describes the number of degrees of freedom associated at a given scale. This leads to an interesting result about the irreversibility of the renormalization group, i.e., along the renormalization group flows, the information about the smaller scale is lost. This result implies that in two dimensions renormalization group flow cannot exhibit a cyclic or chaotic structure. This proof was due to Zamolodchikov, who also showed that at the fixed points in the renormalization group flows, scale invariance is enhanced to conformal invariance.

Noether's theorem states that corresponding to every continuous symmetry, there is an associated conserved current. If the system possesses scale invariance, there is an associated scale current. Similarly, if the system has conformal invariance, there is an associated conserved conformal current. We will demonstrate how the form of these currents imposes constraints on the trace of the stress tensor. We will show that conformal invariance at the classical level implies vanishing trace of the stress tensor. This will provide an intuitive idea of when scale invariance implies conformal invariance. We will explore some examples of scale invariant but not necessarily conformally invariant quantum field theories.

We have a rigorous proof of scale invariance implying conformal invariance due to Zamolodchikov and Polchinski in two dimensions, but as we will show, we also have a counterexample in two dimensions. In the last chapter, we will explore this counterexample due to Hull and Townsend of the nonlinear sigma model. We will demonstrate that for the two-dimensional nonlinear sigma model at one-loop scale invariance does not imply conformal invariance.

The concepts we will explore are specific to two dimensions, and there is no non-perturbative proof that scale invariance implies conformal invariance in greater than two dimensions. This is due to the fact that the conformal group in two dimensions is infinite-dimensional, and the corresponding conformal field theory highly restricts various correlators to have a fixed form, as was shown in the seminal work of Belavin, Polyakov, and Zamolodchikov. In higher dimensions, there are not many restrictions, and the two-dimensional proof does not go through. The proof in two dimensions is only for compact spaces, as this leads to a discrete spectrum of the scaling dimensions, which is essential for the proof of enhancement of scale invariance to conformal invariance.

Chapter 2

Scale And Conformal Invariance In Quantum Field Theory

2.1 Introduction

In this chapter, we will start by showing the mathematical distinction between the scale and conformal transformation. We will argue based on the group structure that scale invariance does not imply conformal invariance. We will then discuss the representations of the conformal group, we will show how the generators of the conformal group act on the fields, and then discuss about unitary highest-weight representation of the conformal group. Then, we will show the implications of this group structure on the energy-momentum tensor and the constraints on the energy-momentum tensor due to scale and conformal invariance. We will discuss the radial quantization and how this allows us to interpret the dilation generator as the Hamiltonian of the system. This will also tell about the discreteness of the spectrum in conformal field theory, which will be useful in the next chapter. We will discuss the consequences of unitarity and its Euclidean counterpart, reflection positivity. Finally, we will show how the conformal invariance constraints the form of two and three-point functions.

2.2 The Conformal Group

In this section, we will provide the mathematical distinction between scale invariance and conformal invariance. We want to understand the relation between scale and conformal invariance and whether the former necessarily implies the latter. The Poincaré algebra is the fundamental symmetry of spacetime in special relativity. Representing the Poincaré algebra in terms of particles leads to the formalism of quantum field theory. The generators of the Poincaré group satisfy the following algebra [3]

$$[J_{\mu\nu}, J_{\rho\sigma}] = i(\eta_{\nu\rho}J_{\mu\sigma} + \eta_{\mu\sigma}J_{\nu\rho} - \eta_{\mu\rho}J_{\nu\sigma} - \eta_{\nu\sigma}J_{\mu\rho}), \quad (2.1)$$

$$[P_\mu, P_\nu] = 0, \quad (2.2)$$

$$[P_\mu, J_{\alpha\beta}] = i(\eta_{\mu\alpha}P_\beta - \eta_{\mu\beta}P_\alpha). \quad (2.3)$$

We can augment the Poincaré algebra by introducing a dilation generator D if the theory under consideration is massless and scale invariant

$$[D, P_\mu] = iP_\mu, \quad (2.4)$$

$$[D, J_{\mu\nu}] = 0. \quad (2.5)$$

The Coleman-Mandula theorem, which says that under a given set of conditions the total symmetry group of the S -matrix is a direct product of the internal symmetry group and the Poincaré group, states that for $d \geq 3$, the maximally enhanced bosonic symmetry of the spacetime for massless particles is achieved by introducing the generators of the special

conformal transformation

$$[D, K_\mu] = -iK_\mu, \quad (2.6)$$

$$[K_\mu, P_\nu] = 2i(\eta_{\mu\nu}D - J_{\mu\nu}), \quad (2.7)$$

$$[K_\rho, L_{\mu\nu}] = i(\eta_{\rho\mu}K_\nu - \eta_{\rho\nu}K_\mu), \quad (2.8)$$

$$[K_\mu, K_\nu] = 0. \quad (2.9)$$

As the dilations arise naturally in the commutation relations involving translations and special conformal transformations. Without scale invariance, the full conformal algebra cannot be realized, meaning the symmetry structure is incomplete. This implies that the conformal invariance requires scale invariance for the closure of the algebra. But on the contrary, the Poincaré group together with dilations forms a subgroup of the full conformal group. This means that a theory invariant under translations, rotations, and dilations is not usually invariant under special conformal transformations. This means that scale invariance does not necessarily imply conformal invariance. However, there are many examples of field theories where we have enhanced conformal invariance rather than just scale invariance.

2.3 Representations of the Conformal Group

A field theory has conformal symmetry at the classical level if the action is invariant under conformal transformation. The conformal invariance in quantum field theories does not follow from the conformal invariance at the classical level. In quantum field theories, we introduce a renormalization condition that introduces a scale in the theory. This scale breaks the conformal symmetry except at the renormalization group fixed point, where the theory becomes scale-independent. The conformal group may be realized as acting on spacetime functions through the differential operator

$$\hat{P}_\mu = -i\partial_\mu \text{ (translation),} \quad (2.10)$$

$$\hat{J}_{\mu\nu} = i(x_\mu\partial_\nu - x_\nu\partial_\mu) \text{ (rotations),} \quad (2.11)$$

$$\hat{D} = -ix^\mu\partial_\mu \text{ (dilations),} \quad (2.12)$$

$$\hat{K}_\mu = -i(2x_\mu x^\nu\partial_\nu - x^2\partial_\mu) \text{ (special conformal transformation).} \quad (2.13)$$

These differential operators act on the Hilbert space in the Schrödinger picture and the local operators in the Heisenberg picture. The action of Poincaré generators on the local fields in the Heisenberg picture is given as

$$[P_\mu, \Phi(x)] = -i\partial_\mu\Phi(x), \quad (2.14)$$

$$[J_{\mu\nu}, \Phi(x)] = i(x_\mu\partial_\nu - x_\nu\partial_\mu)\Phi(x) + S_{\mu\nu}\Phi(x), \quad (2.15)$$

where $S_{\mu\nu}$ is the spin operator associated with the field $\Phi(x)$. To determine the action of the generators of the conformal group on the local fields, we start with the subgroup that leaves the origin $x = 0$ invariant [3]. This subgroup is comprised of rotations, dilations, and special conformal transformations. This smaller subgroup is the same as the augmented Poincaré group with dilations due to the similar roles played by P_μ and K_μ . Then, using the commutation relations of this smaller algebra together with the Hausdorff formula, we get

$$[D, \Phi(x)] = -i(\Delta + x^\mu\partial_\mu)\Phi(x), \quad (2.16)$$

$$[K_\mu, \Phi(x)] = (-2ix_\mu\Delta - x^\nu S_{\mu\nu} - 2ix_\mu x^\nu\partial_\nu + ix^2\partial_\mu)\Phi(x), \quad (2.17)$$

where Δ is the scaling dimension matrix of the collection of the local operators $\Phi(x)$. If we demand that the field $\Phi(x)$ belong to an irreducible representation of the spin group $SO(d -$

1, 1) then any matrix which commutes with all the generators of $S_{\mu\nu}$ must be proportional to identity (Schur's lemma). This means that the scaling dimension matrix Δ will be diagonal and will be proportional to the scaling dimension of the field $\Phi(x)$. In quantum field theory, invariance under the conformal group requires

$$P_\mu|0\rangle = J_{\mu\nu}|0\rangle = K_\mu|0\rangle = D|0\rangle = 0, \quad (2.18)$$

where $|0\rangle$ is the vacuum of the theory under consideration. This states that fields in (2.17) are primary fields, unitary highest-weight representation, if they are annihilated by K_μ at $x = 0$. This vacuum property can be translated to Ward identities in quantum field theories. Now that we have looked at the representation of the conformal group, in the next section, we will show the implications of this on the stress tensor.

2.4 Symmetries and Energy-Momentum Tensor

In the last section, we saw the symmetry of the field theory as an algebra of conserved charges on the Hilbert space. In this section, we will see the implications of the conformal symmetry on the energy-momentum tensor. Let us denote by $g_{\mu\nu}$ the metric tensor in a spacetime of dimension d . A conformal transformation is a diffeomorphism that leaves the metric invariant up to a scale factor

$$g'_{\mu\nu}(x') = \Lambda(x)g_{\mu\nu}(x). \quad (2.19)$$

It is a local change of scale that preserves the angle between two curves. The set of conformal transformations forms a group, and Poincaré group is a subgroup with $\Lambda(x) \equiv 1$. The existence of translational invariance implies that there is a conserved energy-momentum tensor (not generally symmetric)

$$\partial^\mu T_{\mu\nu} = 0. \quad (2.20)$$

If we also have Poincaré invariance, then we can construct a symmetric energy-momentum called the Belinfante stress tensor,

$$T_{\mu\nu} = T_{\nu\mu}. \quad (2.21)$$

The idea behind the construction of the Belinfante stress tensor is that to the canonical energy-momentum tensor, we can add the divergence of a tensor $B^{\rho\mu\nu}$ antisymmetric in the first two indices [3]. Then we can write the canonical conserved current

$$j^{\mu\nu\rho} = T_c^{\mu\nu} x^\rho - T_c^{\mu\rho} x^\nu + \frac{1}{2} i \frac{\partial\Phi}{\partial(\partial_\mu\Phi)} S^{\nu\rho} \Phi, \quad (2.22)$$

as

$$j^{\mu\nu\rho} = T_B^{\mu\nu} x^\rho - T_B^{\mu\rho} x^\nu, \quad (2.23)$$

where $T_B^{\mu\nu}$ is the improved stress tensor called Belinfante stress tensor. The above relation ensures that $T_{\mu\nu} = T_{\nu\mu}$, which can be easily seen by applying the conservation law as long as $\partial_\mu T_B^{\mu\nu} = 0$. This means that the Belinfante tensor is symmetric on-shell.

As we will see in 4 that invariance under scale transformation $x \rightarrow \lambda x$ requires that

$$T^\mu_\mu = -\partial_\mu K^\mu(x). \quad (2.24)$$

This condition ensures the existence of a conserved dilation current $S^\mu(x) = x^\nu T^\mu_\nu + K^\mu(x)$. The first term of the current generates the spacetime dilation, and the second term is associated with the scaling of the fields to preserve the scale invariance of the theory.

Special conformal transformation is a symmetry under

$$x^\mu \rightarrow \frac{x^\mu + b^\mu x^2}{1 - 2b \cdot x + b^2 x^2}. \quad (2.25)$$

This is the finite version of the special conformal transformation, which is generated by the exponential of the infinitesimal version $x^\mu \rightarrow x^\mu + 2(b \cdot x)x^\mu - b^\mu x^2$. There is another way to generate the finite version of the special conformal transformation, which is by realizing that the special conformal transformation is an inversion followed by a translation, followed by another inversion. However, invariance under conformal transformation does not imply invariance under inversion. Inversion is a disconnected piece of the conformal group. As we saw in this section, conformal transformations are diffeomorphisms that leave the metric invariant up to a scale factor. Consider an infinitesimal coordinate transformation $x^\mu \rightarrow x^\mu + \epsilon^\mu(x)$, under such a transformation, the change in metric is

$$g_{\mu\nu} \rightarrow g_{\mu\nu} - (\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu). \quad (2.26)$$

Requiring that this be a conformal transformation implies

$$(\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu) = f(x) g_{\mu\nu}. \quad (2.27)$$

This implies

$$f(x) = \frac{2}{d} \partial_\mu \epsilon^\mu. \quad (2.28)$$

The variation of action under such transformation is given by

$$\begin{aligned} \delta S &= \int d^d x T^{\mu\nu}(x) \partial_\mu \epsilon_\nu \\ &= \frac{1}{2} \int d^d x T^{\mu\nu}(x) (\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu) \\ &= \frac{1}{d} \int d^d x T^\mu_\mu(x) \partial_\rho \epsilon^\rho. \end{aligned}$$

Thus, the condition for conformal invariance is that the energy-momentum tensor be traceless. The converse is not true because $\partial_\mu \epsilon^\mu$ is not an arbitrary function. The condition for conformal invariance is

$$T^\mu_\mu = 0. \quad (2.29)$$

We will derive the form of the conformal current in chapter (4).

2.5 Unitarity

We have already seen from the group structure that there is no reason for scale invariance to imply conformal invariance. In the previous section, we showed how these properties can be stated in terms of the energy-momentum tensor. In this section, we look at the implications of unitarity and discreteness of the spectrum in quantum field theories.

We will first begin with a discussion of radial quantization. Hilbert spaces and operators are of importance in conformal field theory because of the power of the algebraic and group-theoretical methods. The operator formalism requires a choice of reference frame, as it is not manifestly Lorentz invariant, which means we have to choose a direction of the time axis. The choice of time direction is arbitrary in Euclidean field theories, we can choose the radial direction to be the direction of time. As we will show that this choice allows us to reinterpret the dilation transformation as time translation in the complex plane. As discussed in [3], we begin with flat Euclidean space and time coordinates σ^1 and σ^0 . In Minkowski space, the

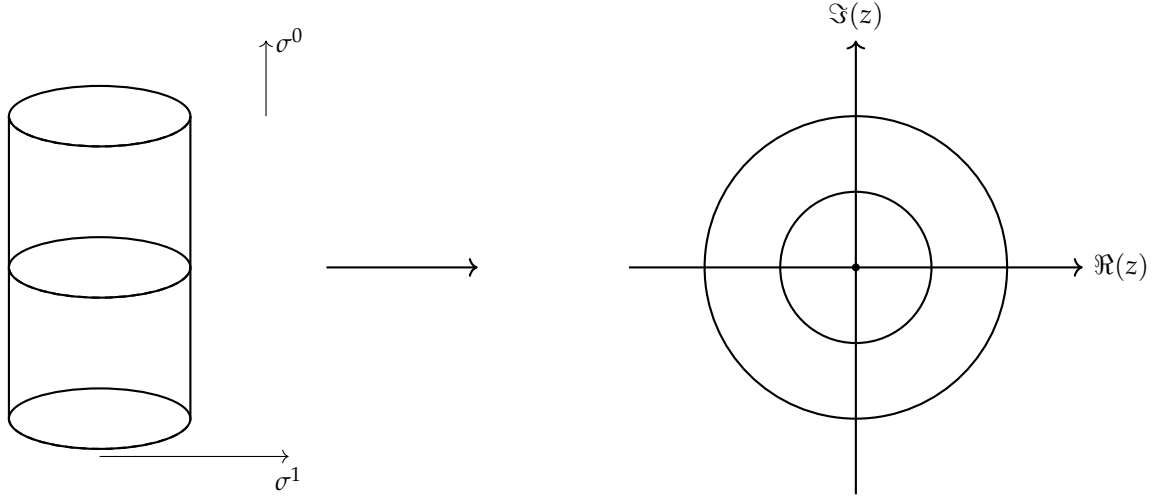


FIGURE 2.1: Map of cylinder onto a plane

standard light cone coordinates are $\sigma^0 \pm \sigma^1$, in Euclidean space the corresponding analogues are $\zeta, \bar{\zeta} = \sigma^0 \pm i\sigma^1$. To eliminate infrared divergences [6], we compactify the space coordinate, $\sigma^1 \equiv \sigma^1 + 2\pi$. This compactification defines a cylinder with coordinates σ^1, σ^0 . Consider the conformal map

$$\zeta \rightarrow z = \exp \zeta \quad (2.30)$$

that maps the cylinder to the complex plane (2.1). The time direction is along the axis of the cylinder. When mapped to the complex plane, the time direction is radial. The dilations, $z \rightarrow \exp(a)z$ on the cylinder correspond to time translations in the complex plane, and therefore we can identify the generator of dilation can be regarded as being proportional to the Hamiltonian. This process of defining quantum field theory on a plane is called radial quantization [3]. The notion of the discreteness of the spectrum comes from the radial quantization. A conformal field theory has a discrete spectrum means that the spectrum under the time evolution is discrete, this is because in the original theory, the eigenvalues of the dilation generator are discrete.

We have chosen the symmetry algebra to be represented by unitary operators acting on the Hilbert space. The unitary representation of the Poincaré algebra provides the spectral decomposition of the two-point function [11]. The general representation of the Feynman two-point function for a scalar operator $\phi(x)$ satisfies the Källén-Lehmann spectral decomposition

$$\langle \Omega | \phi(x) \phi(y) | \Omega \rangle = \int_0^\infty \frac{dM^2}{2\pi} \rho(M^2) \int \frac{d^d k}{2\pi} \frac{i \exp(-ik(x-y))}{k^2 - M^2 + i\epsilon}, \quad (2.31)$$

with a positive definite spectral density, because the Hilbert space has a positive definite norm. We know that the under scale transformations $x \rightarrow \lambda x$ the fields $\phi(x)$, with scaling dimension Δ , transform as $\phi(x) \rightarrow \lambda^{-\Delta} \phi(x)$. Using the above equation, it tells us that scale invariance requires the spectral function to have the following form under scale transformations

$$\rho(M^2) \propto (M^2)^{\Delta-d/2}. \quad (2.32)$$

Another important aspect is the unitarity of the S -matrix

$$S^\dagger S = \mathbb{1}. \quad (2.33)$$

The S -matrix does not exist for scale and conformally invariant field theories. This is because, unlike in typical quantum field theories, the two-point function decays as a power law in scale and conformally invariant field theories, which in momentum space has a branch cut

rather than an isolated pole. This implies there is no discrete one-particle state. Conformal transformations map points far away to nearby points, so there is no clear distinction between the interaction region and asymptotically far-away points. This can lead to IR-divergences and ill-defined overlap integrals. As I will be dealing mostly with Euclidean field theories, after Wick rotation, the statement of unitarity is equivalent to reflection positivity. Reflection positivity states

$$\langle \phi(x) \Theta[\phi(x)] \rangle \geq 0, \quad (2.34)$$

where $\Theta[\phi(x_1, \dots, x_d)] = \phi^\dagger(x_1, \dots, -x_d)$ is the Euclidean adjoint. Reflection positivity will play a very important role in the proof of enhancement of scale invariance to conformal invariance in $d = 2$ in chapter 3.

2.6 Consequence of Scale and Conformal Invariance

Scale and conformal invariance severely constrain the form of the correlation functions. Conformal symmetry is powerful because it constrains correlation functions to specific functional forms. These forms are entirely determined (up to constants) by the symmetries and scaling dimensions of the operators involved. All correlation functions must scale in accordance with the scaling dimension matrix as dictated by the Ward-Takahashi identity for dilation. The scalar two-point function of a scale-invariant theory is given by

$$\langle \phi_1(x) \phi_2(y) \rangle = \frac{C_{12}}{|x - y|^{\Delta_1 + \Delta_2}}. \quad (2.35)$$

Conformal invariance constrains it even more by requiring $\Delta_1 = \Delta_2$

$$\langle \phi_1(x) \phi_2(y) \rangle = \frac{C_{12} \delta_{\Delta_1, \Delta_2}}{|x - y|^{2\Delta_1}}. \quad (2.36)$$

The reason behind the above constraint is that, as we can verify from (2.25) that for such a transformation

$$\left| \frac{\partial x'}{\partial x} \right| = \frac{1}{1 - 2b \cdot x + b^2 x^2}. \quad (2.37)$$

Using this we can easily verify that unless $\Delta_1 = \Delta_2$ (2.35) will not be conformally invariant. Conformal invariance also fixes the form of the three-point function to be [3]

$$\langle \phi_1(x_1) \phi_2(x_2) \phi_3(x_3) \rangle = \frac{C_{123}}{x_{12}^{\Delta_1 + \Delta_2 - \Delta_3} x_{23}^{\Delta_3 + \Delta_2 - \Delta_1} x_{13}^{\Delta_1 + \Delta_3 - \Delta_2}}. \quad (2.38)$$

There are many more interesting results involving conformal invariance of the theory, like the four-point function in conformally invariant field theories satisfy the bootstrap equations [3]. The bootstrap hypothesis is the only dynamical input necessary to solve the theory completely. While the bootstrap framework offers a powerful, symmetry-based method for solving conformal field theories, our focus will remain on the scale-to-conformal enhancement and associated renormalization group structure.

In this chapter, we demonstrated the mathematical distinction between scale and conformal invariance and how, from the group structure, we understand that scale invariance does not imply conformal invariance. We studied the representations of the conformal group and how the fields transform under the action of the generators of the conformal group. As demonstrated, the form of the current and the conditions on the stress tensor under scale and conformal invariance. This translated the ideas about the group structure into implications on the energy-momentum tensor. We then explored the implications of unitarity and how this idea translates to Euclidean field theories, called reflection positivity. Radial quantization told

us about the discreteness of the spectrum and why the dilation generator is regarded as being proportional to the Hamiltonian of the system. These results will be used extensively in the following chapters to understand the conditions under which we have enhanced conformal invariance from scale invariance.

Chapter 3

C-theorem

3.1 Introduction

In this chapter, we will explore the well-established proof in two dimensions where the scale invariance is enhanced to conformal invariance. In [17] it was shown that under the assumptions of renormalizability, unitarity and Poincaré invariance, scale invariance is enhanced to conformal invariance in two-dimensional field theories. We can learn about many interesting properties of the theory by studying it at a fixed point. The field theory at the fixed point has a new emergent symmetry, the conformal symmetry. The full analysis of the theory at a fixed point involves constructing the conformally invariant field theory, i.e., determining the central charge and structure constants of the three-point function, and then analyzing the behavior of the renormalization group near the fixed point. The existence of a symmetry in quantum field theory gives rise to Ward identities, the quantum analogue of Noether's theorem. At a fixed point in two-dimensional field theory, the global conformal group is isomorphic to $SL(2, \mathbb{C})$, this is enhanced to an infinite-dimensional Virasoro algebra once we impose local conformal invariance. The existence of the infinite-dimensional algebra imposes a lot of constraints on the system, which are incorporated through Ward identities. This simplifies the analysis and leads to an exact solution of the theory based on symmetry arguments alone. If the solution corresponding to the fixed points is known exactly, the properties of the renormalization group in the vicinity of the fixed point can be determined using perturbation theory. The perturbation theory only applies if the renormalization group shows some nontrivial topological behavior in a small neighborhood of the fixed point. We will see that such properties appear if the conformal dimension is close to unity. We will use this approach to calculate properties of the renormalization group using perturbation theory about the fixed point. This is the idea behind the ϵ -expansion.

In 3.3, we will show that under the assumptions of positivity and renormalizability, the renormalization group possesses the following properties

1. There exists a function $c(g) \geq 0$ such that we have

$$\frac{d}{d\tau} c \equiv \sum_i \beta^i(g) \frac{\partial}{\partial g^i} c(g) \leq 0. \quad (3.1)$$

The equality is reached only at the fixed point.

2. At the fixed point $c(g)$ is stationary, i.e.,

$$\beta^i(g) = 0 \implies \frac{\partial c(g)}{\partial g^i} = 0. \quad (3.2)$$

3. At the fixed point, the value of $c(g)$ is equal to the central charge of the Virasoro algebra.

We will compute in 3.3 the renormalization group β -functions in power series expansion about the fixed point up to second order in coupling constants and $c(g)$ up to third order in

coupling constant, and show that to this order the following relation holds

$$\beta^i(g) = -\frac{1}{12}G^{ij}(g)\frac{\partial c(g)}{\partial g^j}, \quad (3.3)$$

where $G_{ij}(g)$ is a symmetric positive-definite matrix which specifies the metric in the space of coupling constants.

The "c-theorem" has an obvious consequence, as we flow from UV to IR under the action of the renormalization group flows and let the conformal field theory corresponding to these two points have the central charge c_{UV} and c_{IR} respectively then

$$c_{UV} > c_{IR}. \quad (3.4)$$

This gives meaning to the renormalization group as the ordering of the conformal field theory solutions by the magnitude of the central charge c . The "c-theorem" rigorously shows that the renormalization group "time" has a preferred direction in an interacting quantum field theory [17]. This means the motion in the space of the coupling constant under the renormalization group becomes irreversible. This rules out the possibilities like limit cycles in two-dimensional quantum field theories.

3.2 The Renormalization Group in Two-Dimensional Field Theory

The renormalization group is a powerful framework to analyze the qualitative properties of quantum field theories. In quantum field theory, when we encounter a divergent loop integral in perturbation theory, we use different methods like imposing a sharp cutoff as the upper limit in the integral, but this method is not very useful in theories like QED, as it violates the Ward identity. So we try different approaches like Pauli-Villars regularization in which we subtract out from the photon propagator the contribution due to a virtual massive photon but this method although useful cannot explain the physical intuition for introducing a massive virtual photon and the dimensional regularization, in this regularization technique we evaluate the divergent integral in d dimensions and in the final expression for the observable quantity should have well-defined limit as $d \rightarrow 4$. In all of the techniques mentioned above, we impose a cutoff by hand, but Wilson's analysis [16] takes a different point of view, that any quantum field theory is defined fundamentally with a cutoff Λ that has some physical significance. In statistical mechanics, the momentum cutoff is defined as the inverse of the atomic spacing. In QED and other quantum field theories, the cutoff can be associated with some fundamental graininess of the spacetime, perhaps as a result of quantum fluctuations in gravity. The procedure for determining the renormalization group can be summarized as follows: We integrate out the high-momentum modes, $t\Lambda < k < \Lambda$, where t is a number between $[0, 1]$. We then rescale the new momentum $k' = \frac{k}{t}$ and finally rescale the fields so that the gradient term remains canonically renormalized.

We denote by $S(g, \Lambda)$ the Euclidean action of a field theory, $S = \int d^d x \mathcal{L}(g, \Lambda, x)$, where Λ is the ultraviolet cutoff and the functional also depends on a (possibly infinite) set of parameters $g = (g^1, g^2, \dots)$, which are called coupling constants. Then the transformation defined in the previous paragraph define a single-parameter group of motions in the space (Q) of coupling constants g , $R_t Q \rightarrow Q$ which gives us an effective action such that the correlation function of the original theory and the theory described effective action agree at small momentum scales ($k \ll \Lambda$). It is the study of these flows in the parameter space (Q) which tells us about physical problems, and is the essence of the renormalization group. The components of the vector field that generate these flows are called the " β -functions":

$$dg^i = \beta^i(g)dt. \quad (3.5)$$

The β -functions tell us how the coupling constants change under the flow generated by the renormalization group transformation. As we integrate out the heavy momentum modes, some of the information about the ultraviolet behavior is lost under the renormalization group transformation. There is no way to reverse this process, so the motion in the parameter space Q will become an irreversible process. The main object of interest in quantum field theories is the correlation functions, which we can relate to observables like the S -matrix. The behavior of the correlation functions under the change of scale is described by the Callan-Symanzik equation. In the next two sections we will derive the Callan-Symanzik equation in two dimensions using two different concepts, one under the local renormalization group transformation which realizes the infinitesimal Weyl rescaling of the metric and is the extension of the usual Callan-Symanzik equation and the other under the renormalization group transformation as discussed above (homogeneous renormalization group). From here on now we will be dealing with two-dimensional Euclidean field theory.

3.2.1 Homogeneous Renormalization Group

A quantum field theory can be regarded as the set of correlation functions of local fields $A_i(x)$ defined as,

$$\langle A_1(x_1) \dots A_n(x_n) \rangle = \int [\mathcal{D}\phi] (A_1(x_1) \dots A_n(x_n)) \exp(-S[\phi]), \quad (3.6)$$

where ϕ is a set of "fundamental fields" and $S[\phi] = \int d^2x \mathcal{L}(\phi(x), \partial_\mu \phi(x))$ is the local action. I will assume that the correlation function (3.6) is renormalized. The constant term (independent of ϕ) is absorbed in the definition of $S[\phi]$ so that the distribution function is properly normalized. The local fields $A_i(x)$ are composite fields which depend on the fundamental fields ϕ and are elements of infinite dimensional vector space $\mathcal{A}$ and form a closed (commutative and associative) algebra under operator product expansion (OPE):

$$A_i(x_1) A_j(x_2) = \sum_k C_{ij}^k(x_1, x_2) A_k(x_2), \quad (3.7)$$

where $C_{ij}^k(x_1, x_2)$ are complex valued coefficients, called structure constants. In terms of the OPE coefficients, the associativity is expressed as follows:

$$\sum_t C_{ij}^t(x_1, x_2) C_{it}^m(x_2, x_3) = \sum_l C_{il}^m(x_1, x_3) C_{jk}^l(x_2, x_3). \quad (3.8)$$

The information about spatial symmetries of a homogeneous and isotropic theory (invariance under the full Euclidean group) is realized by the existence of a symmetric stress tensor $T_{\mu\nu}(x) \in \mathcal{A}$ satisfying,

$$\partial_\mu T^{\mu\nu}(x) = 0. \quad (3.9)$$

The components of the stress tensor describe the variation of Lagrangian density $(\mathcal{L})(\phi, \partial_\mu \phi)$ under infinitesimal coordinate transformation $x_\mu \rightarrow x_\mu + \delta x_\mu$. Let us consider the case of variation of $\mathcal{L}(x) \equiv \mathcal{L}(\phi(x), \partial_\mu \phi)$ under scale transformation

$$x_\mu \rightarrow x_\mu + \frac{1}{2} dt x_\mu. \quad (3.10)$$

Under a general infinitesimal change of coordinates $x_\mu \rightarrow x_\mu + \epsilon_\mu(x)$, the induced change in the action is

$$\delta S = \int d^2x T^{\mu\nu} \partial_\mu \epsilon_\nu(x), \quad (3.11)$$

under a local change of scale $\epsilon_\nu = x_\nu dt$, and we get

$$\mathcal{L}(x) \rightarrow \mathcal{L}(x) - dt\Theta(x), \quad (3.12)$$

where $\Theta(x) = -T_\mu^\mu(x)$. Now we assume that there is an n -parameter family of field theories, meaning the correlation function in (3.6) depends on coupling constants, $g^i = (g^1, g^2, \dots, g^n)$. Now, since the theory under consideration is Euclidean invariant theory, i.e., it is homogeneous and isotropic due to translational and rotational invariance, implying that the $\mathcal{L}(x)$ is a scalar under these symmetries. If we have a family of actions defined as

$$S[x, \{g^i\}] = \int d^2x [\mathcal{L}(x) + \sum_i g^i \Phi_i(x)], \quad (3.13)$$

then by definition

$$\Phi_i(x) = \frac{\partial \mathcal{L}(x)}{\partial g^i} \quad (3.14)$$

is a spinless field, $\Phi_i \in \mathcal{A}^{(0)}$ and linear operators $\mathcal{B}_i : \mathcal{A}^s \rightarrow \mathcal{A}^s$ such that from (3.6) it follows that

$$\frac{\partial}{\partial g^i} \langle A_1(x_1) \dots A_N(x_N) \rangle = \sum_{a=1}^N \langle \mathcal{B}_{i,a} A_1(x_1) \dots A_N(x_N) \rangle + \int d^2y \langle A_1(x_1) \dots A_n(x_n) \Phi_i(y) \rangle, \quad (3.15)$$

where the operator $\mathcal{B}_{i,a}$ acts on $A_a(x_a)$ as

$$\mathcal{B}_i A(x) = \frac{\partial A(x)}{\partial g^i}. \quad (3.16)$$

The operator $\mathcal{B}_i$ takes into account explicit dependence of the local fields on the coupling constants, which is necessary when the integral in (3.15) is divergent as $y \rightarrow x_i$. The subspace $\Phi \in \mathcal{A}^{(0)}$ spanned by the fields Φ_i is associated with the tangent space $T_g Q$, where Q is the parameter space or the space of all coupling constants with coordinates g^i . If the field theories under consideration are renormalizable, we have the operator relation

$$\Theta(x) = \sum_i^n \beta_i(g) \Phi_i(x), \quad (3.17)$$

That is for all $g \in Q$ the field $\Theta(x) \in \Phi$. I will derive this relation in the next section when considering the local renormalization group. We have already seen the variation of Lagrangian density under a local change of scale in (3.12). Using (3.6), we get the relation

$$\sum_{a=1}^N \left\langle \left(\frac{1}{2} x_a^\mu \frac{\partial}{\partial x_a^\mu} + \mathcal{D} \right) A_1(x_1) \dots A_n(x_n) \right\rangle = \int d^2y \langle A_1(x_1) \dots A_n(x_n) \Theta(y) \rangle. \quad (3.18)$$

The operator $\mathcal{D} : \mathcal{A}^{(s)} \rightarrow \mathcal{A}^{(s)}$ describes the variation of the fields $A \in \mathcal{A}$ under the infinitesimal scale transformation (3.10):

$$A(0) \rightarrow A(0) + dt \mathcal{D} A(0). \quad (3.19)$$

The integral in (3.18) can diverge as $y \rightarrow x_i$, in this case the operator $\mathcal{D}$ must depend on the cutoff in such a way to cancel the divergence since the correlation functions are renormalized (they do not depend on cutoff). The same argument extends to the operator $\mathcal{B}_i$. Multiply

(3.15) with $\beta_i(g)$ and sum over all i together with (3.17) to get

$$\sum_{i=1}^n \beta^i(g) \frac{\partial}{\partial g^i} \langle A_1(x_1) \dots A_N(x_N) \rangle = \sum_{i=1}^n \sum_{a=1}^N \langle \beta^i(g) \mathcal{B}_{i,a} A_1(x_1) \dots A_N(x_N) \rangle + \int d^2y \langle A_1(x_1) \dots A_n(x_n) \Theta(y) \rangle. \quad (3.20)$$

Using (3.18) and (3.20) we get the Callan-Symanzik equation

$$\left[\sum_{a=1}^N \left(\frac{1}{2} x_a^\mu \frac{\partial}{\partial x_a^\mu} + \Gamma_a(g) \right) - \sum_{i=1}^n \beta^i(g) \frac{\partial}{\partial g^i} \right] \langle A_1(x_1) \dots A_N(x_N) \rangle = 0, \quad (3.21)$$

where the operator Γ_a is defined as follows

$$\Gamma = \mathcal{D} + \beta^i(g) \mathcal{B}_i. \quad (3.22)$$

This is the matrix of anomalous dimension. This matrix tells us how the operator's dimensions change with the scale change. The Callan-Symanzik equation (3.21) tells us how the renormalized correlation function changes with a change in energy scale. It tells us that in parameter space Q , field theories that correspond to different points of the same curve, which are solutions of (3.5), differ only by a scale transformation. The Callan-Symanzik equation also tells us that the subspace Φ is invariant under the action of the matrix of anomalous dimension, this can be thought of as this way, $\Phi_i \in \Phi$ span the tangent space $T_g Q$ and β^i is a dual vector, their contraction $\Theta(x) = \sum_i \beta^i(g) \Phi_i(x)$ should not change under mere change of basis. This means if there is a non-trivial mixing of these operators, the operator relation (3.17) will be violated. This can be seen by considering the equations (3.12), (3.19), (3.17), and (3.14) as follows, under scaling, the change in Lagrangian is given by (3.12), and so we have a set of new fields defined as

$$\Phi'_j = \frac{\partial \mathcal{L}'}{\partial g^j} = \Phi_j - dt \sum_i \frac{\partial \beta^i}{\partial g^j} \Phi_i - dt \sum_i \beta^i \frac{\partial \Phi_i}{\partial g^j}. \quad (3.23)$$

And according to (3.19) the change in field Φ_j is given as

$$\Phi'_j = \Phi_j + dt \mathcal{D} \Phi_j. \quad (3.24)$$

The above two equations must be the same as, then we get

$$\begin{aligned} dt \mathcal{D} \Phi_j &= -dt \sum_i \frac{\partial \beta^i}{\partial g^j} \Phi_i - dt \sum_i \beta^i \frac{\partial \Phi_i}{\partial g^j} \\ &\Rightarrow \mathcal{D} \Phi_j + \sum_i \beta^i \frac{\partial \Phi_i}{\partial g^j} = - \sum_i \frac{\partial \beta^i}{\partial g^j} \Phi_i \\ &\Rightarrow \left(\delta_j^i + \mathcal{D} \delta_j^i + \sum_i \beta^i \frac{\partial}{\partial g^j} \right) \Phi_i = \left(\delta_j^i - \sum_i \frac{\partial \beta^i}{\partial g^j} \right) \Phi_i \\ &\Rightarrow \gamma_j^i(g) \Phi_i = \left(\delta_j^i - \sum_i \frac{\partial \beta^i}{\partial g^j} \right) \Phi_i. \end{aligned} \quad (3.25)$$

This relation shows explicitly that the subspace $\Phi \in \mathcal{A}^0$ is invariant for the matrix of anomalous dimension, that is

$$\Gamma \Phi_i = \gamma_i^j(g) \Phi_j = \left(\delta_i^j - \frac{\partial \beta^j}{\partial g^i} \right) \Phi_j. \quad (3.26)$$

This relation is equivalent to the important statement that the renormalizations of the stress tensor components are absent, i.e.,

$$(\Gamma - 1)T^{\mu\nu} = 0. \quad (3.27)$$

This can be derived as follows, using (3.22)

$$\begin{aligned} \Gamma\Theta &= \left(\mathcal{D} + \beta^i \frac{\partial}{\partial g^i} \right) \beta^j \Phi_j \\ &= \mathcal{D}\beta^j \Phi_j + \beta^i \beta^j \frac{\partial \Phi_j}{\partial g^i} + \beta^i \frac{\partial \beta^j}{\partial g^i} \Phi_j \\ &= \beta^j \Gamma \Phi_j + \beta^i \frac{\partial \beta^j}{\partial g^i} \Phi_j. \end{aligned}$$

Now using (3.26) we get

$$(\Gamma - 1)\Theta = 0. \quad (3.28)$$

The equation for the stress tensor in complex coordinates takes the form

$$\partial_z T = \partial_z \Theta, \quad \partial_z \bar{T} = \partial_z \bar{\Theta}. \quad (3.29)$$

The action of (3.22) on (3.29) equations and using the fact that coordinates commute with (3.22) we get

$$(\Gamma - 1)T = (\Gamma - 1)\bar{T} = 0. \quad (3.30)$$

This tells us that the stress tensor receives no rescaling and no anomalous dimension. This shows that the stress tensor as a composite operator is an eigenvector of the matrix of anomalous dimension with eigenvalue one. It reflects the fact that the stress tensor is protected by symmetry, it generates space-time translations, and its scaling is fixed by conservation and symmetry arguments.

3.2.2 Local Renormalization Group

In the last section, we stated without proof that if the field theory under consideration is renormalizable, then (3.17) holds as an operator equation. The discussion in this section follows from [10]. A local renormalization group describes how the correlation functions change under the effect of local Weyl rescaling of the metric. For a field theory, the physical observables can be expressed in terms of the Schwinger functional W defined as

$$\exp(-W[g]) = \int [\mathcal{D}\phi] \exp(-S), \quad (3.31)$$

where S is formed from the bare action depending on cutoff Λ , such that $W[g]$ is finite scalar function of g^i as $\Lambda \rightarrow \infty$. By absorbing all the scale dependence into a redefinition of coupling constants and wave function renormalization, we can make W invariant along the renormalization group flow so that it obeys the Callan-Symanzik equation (3.21),

$$\hat{D}W[g] = 0. \quad (3.32)$$

This simply means that once we run the coupling constants and fields renormalizations appropriately with scale, the physics described by $W[g]$ does not depend on the arbitrary choice of renormalization scale. In this section, I will consider the extension of the homogeneous renormalization group to describe how the correlation functions change under local rescaling of length, which is realized by Weyl transformation, $g_{\mu\nu}(x) \rightarrow \exp(-2\sigma(x))g_{\mu\nu}(x)$, where $\sigma(x)$ is arbitrary smooth function when the field theory is described on a curved background.

These Weyl transformations form an infinite-dimensional Abelian group. As $W[g]$ is a finite scalar functional, we can define the expectation value of the energy-momentum tensor $T_{\mu\nu}(x)$ by functional differentiation

$$T_{\mu\nu}(x) = \frac{4\pi}{\sqrt{g}} \frac{\delta}{\delta g^{\mu\nu}(x)} W. \quad (3.33)$$

We also need to be able to describe a well-defined local operator equations, for this we have to define a set of local operator Φ_i corresponding to a set of renormalizable couplings and which form a closed set under mixing, we saw this argument in the previous section. To achieve this, we have to promote the coupling constants g^i to background position-dependent fields $g^i(x)$ [10] so they act as source terms for the local operators Φ_i , which can be used to define connected correlators. This is necessary to analyze local renormalization group flows and tells us how the couplings will change under a spacetime-dependent Weyl transformation. Translational invariance is lost in this formalism, and to maintain renormalizability, we need to introduce counterterms like $\partial_\mu g^i$. We can therefore define

$$\langle \mathcal{O}_i(x) \mathcal{O}_j(y) \dots \rangle = \alpha \frac{\delta}{\delta g^i(x)} \alpha \frac{\delta}{\delta g^j(y)} W, \quad (3.34)$$

where α is some constant. In this framework, the generator for infinitesimal Weyl transformations $\delta g^{\mu\nu} = 2\sigma g^{\mu\nu}$ is

$$\Delta_\sigma^W = 2 \int d^d x \sqrt{g} \sigma g^{\mu\nu} \frac{\delta}{\delta g^{\mu\nu}}, \quad (3.35)$$

as the couplings are now position dependent, the corresponding variations in the couplings, as determined by the β -function, are given as

$$\Delta_\sigma^\beta = \int d^d x \sqrt{g} \sigma \frac{\delta}{\delta g^i(x)}. \quad (3.36)$$

The local renormalization group equation takes the form

$$\Delta_\sigma^W W = \Delta_\sigma^\beta W + \text{terms involving derivatives of } g_{\mu\nu}, g^i, \sigma. \quad (3.37)$$

This gives us

$$g^{\mu\nu} T_{\mu\nu} = \beta^i \mathcal{O}_i + \text{curvature terms}. \quad (3.38)$$

The additional terms in (3.37) and (3.38) can be calculated using the renormalized perturbation theory. This proves the operator relation (3.17) stated in the previous section.

3.3 Perturbation Theory About Fixed Point

In 3.2.1, we derived the homogeneous Callan-Symanzik equation (3.21), and we showed that the components of the vector field $\beta_i(g)$ generate the flow in the space of coupling constants Q . The global topological properties of $\beta_i(g)$ are of special interest as they help us determine the phase structure of the field theory. This means that if we start from a UV theory, following the renormalization group trajectory, what are the possible different end points we can reach, depending on tuning the initial values of g^i . Assuming that the field theory under consideration satisfies the positivity condition, we will show that in two dimensions, the renormalization group flows exhibit a dissipative behavior. For this, consider two-point

functions

$$\frac{F(\tau)}{z^4} = \langle T(z, \bar{z}) T(0, 0) \rangle, \quad (3.39)$$

$$\frac{H(\tau)}{z^3 \bar{z}} = \langle T(z, \bar{z}) \Theta(0, 0) \rangle, \quad (3.40)$$

$$\frac{G(\tau)}{z^2 \bar{z}^2} = \langle \Theta(z, \bar{z}) \Theta(0, 0) \rangle. \quad (3.41)$$

The argument of the functions $F(\tau)$, $H(\tau)$ and $G(\tau)$ is $\tau = \log(z\bar{z})$ with $z = x_1 + ix_2$ and $\bar{z} = x_1 - ix_2$, z and $\bar{z}$ are treated as independent components. The components of the energy-momentum tensor in this coordinate system are defined as

$$T = T_{11} - T_{22} + 2iT_{12}, \quad \bar{T} = T_{11} - T_{22} - 2iT_{12}. \quad (3.42)$$

Using the conservation of the energy-momentum tensor (3.29), we have the Ward identity

$$g^{\mu\alpha} \partial_\alpha \langle T_{\mu\nu}(x) \chi(y) \rangle = 0. \quad (3.43)$$

The above relation holds only when the insertion point of the field χ does not coincide with the point x . The above equation in complex coordinates takes the form

$$\bar{\partial} \langle T_{z\nu}(z, \bar{z}) \chi \rangle + \partial \langle T_{\bar{z}\nu}(z, \bar{z}) \chi \rangle = 0. \quad (3.44)$$

Now, consider the case $\nu = z$ and $\chi = T(0, 0)$, this gives us

$$\bar{\partial} \langle T(z, \bar{z}) T(0, 0) \rangle - \partial \langle \Theta(z, \bar{z}) T(0, 0) \rangle = 0, \quad (3.45)$$

where we have used $T_{zz} = T$ and $T_{z\bar{z}} = -\Theta$. Now using the definition of functions from (3.41) we get

$$\begin{aligned} & \bar{\partial} \left(\frac{F(\tau)}{z^4} \right) - \partial \left(\frac{H(\tau)}{z^3 \bar{z}} \right) = 0 \\ \Rightarrow & \frac{1}{z^4} \frac{\partial \tau}{\partial \bar{z}} \frac{dF(\tau)}{d\tau} + \frac{3H(\tau)}{z^4 \bar{z}} - \frac{1}{z^3 \bar{z}} \frac{\partial \tau}{\partial z} \frac{dH(\tau)}{d\tau} = 0 \\ \Rightarrow & \frac{1}{z^4 \bar{z}} \dot{F}(\tau) + \frac{3}{z^4 \bar{z}} H(\tau) - \frac{1}{z^4 \bar{z}} \dot{H}(\tau) = 0. \end{aligned}$$

This relation gives us

$$\dot{F} = \dot{H} - 3H. \quad (3.46)$$

The dots denote derivative with respect to τ . Now, consider the case $\nu = z$ and $\chi = \Theta(0, 0)$, this gives us

$$\bar{\partial} \langle T(z, \bar{z}) \Theta(0, 0) \rangle - \partial \langle \Theta(z, \bar{z}) \Theta(0, 0) \rangle = 0. \quad (3.47)$$

Using the definition from (3.41) we get,

$$\bar{\partial} \left(\frac{H(\tau)}{z^3 \bar{z}} \right) - \partial \left(\frac{G(\tau)}{z^2 \bar{z}^2} \right) = 0. \quad (3.48)$$

This gives us

$$\dot{H} - H = \dot{G} - 2G. \quad (3.49)$$

We define a new function

$$c = 2F + 4H - 6G, \quad (3.50)$$

consider the derivative of the above function with respect to τ ,

$$\dot{c} = 2\dot{F} + 4\dot{H} - 6\dot{G}. \quad (3.51)$$

Now, using the relations (3.46) and (3.49) we substitute for $\dot{F}$ and we get

$$\begin{aligned} \dot{c} &= 2\dot{H} - 6H + 4\dot{H} - 6\dot{G} \\ &= 6(\dot{H} - H) - 6\dot{G} \\ &= 6\dot{G} - 12G - 6\dot{G} \\ &\Rightarrow \dot{c} = -12G. \end{aligned} \quad (3.52)$$

This shows that $c(\tau)$ is a monotonically decreasing function of τ as $G(\tau) \geq 0$ by positivity condition. This result shows that $c(g)$ is monotonically decreasing along the renormalization group flow, this function provides a measure of the number of degrees of freedom, which decreases as we flow from UV to IR. If we fix τ , say to $\tau = 0$ (i.e. we fix an energy scale) then the two-point functions F , H and G only depend on the coupling constant g . If we fix an energy scale $\tau = \mu^{-1}$, then the c -function depends on the energy scale μ , but it is not clear that it will depend on coupling constants alone, $c \equiv c((g^i(\mu)))$, when evaluated at energy scale μ and not on μ explicitly [9]. The reason that the dependence is only through $g^i(\mu)$ is renormalizability, we know that the renormalized two-point function does not depend on the renormalization scale M , we obtain the Callan-Symanzik equation for a general correlation function of the energy-momentum tensor, as there is no anomalous dimension of the stress tensor (3.27),

$$\frac{d}{d \log M} \langle T_{\mu\nu}(x) \dots \rangle = \left(\frac{\partial}{\partial \log M} + \beta^i(g) \frac{\partial}{\partial g^i} \right) \langle T_{\mu\nu}(x) \dots \rangle = 0. \quad (3.53)$$

The c -function, as argued above, depends only on the energy scale μ and at $\tau = \mu^{-1}$ is a dimensionless quantity, we get

$$\left(\frac{\partial}{\partial \log \mu} + \frac{\partial}{\partial \log M} \right) c = 0. \quad (3.54)$$

This gives us the chain rule

$$\frac{\partial}{\partial \log \mu} c = \beta^i(g) \frac{\partial}{\partial g^i} c, \quad (3.55)$$

with coupling constants $g^i \equiv g^i(\mu)$. This tells us that at a particular energy scale, the c -function depends only on the coupling constants g^i and has no explicit dependence on μ . Using the above chain rule and (3.17) we obtain

$$\beta^i \frac{\partial}{\partial g^i} c = -12\beta^i(g)\beta^j(g)G_{ij}(g), \quad (3.56)$$

where

$$G_{ij}(g) \equiv G_{ij}(0, g), \quad G_{ij}(\tau, g) = z^2 \bar{z}^2 \langle \Phi_i(z, \bar{z}) \Phi_j(0, 0) \rangle. \quad (3.57)$$

The quantity $ds^2 = G_{ij}(g) dg^i dg^j$ can be regarded as the metric in the space of coupling constants Q [18]. This metric provides a notion of distance on the interaction space and allows us to view renormalization group flows as the gradient of the c -function. Positivity of the metric reflects unitarity. The renormalization group fixed points are determined as the solution of the equation

$$\beta^i(g_{*i}) = 0, \quad i = 1, 2, \dots, n. \quad (3.58)$$

For $g = g_{*l}$, $\Theta = 0$, we have conformal symmetry. This implies that

$$T = T(z), \quad \bar{T} = \bar{T}(\bar{z}). \quad (3.59)$$

We can introduce an infinite set of operators $L_n, \bar{L}_n$ with $n = 0, \pm 1, \pm 2, \dots$. The operators L_n and $\bar{L}_n$ satisfy the following commutation relations

$$[L_n, L_m] = (n - m)L_{n+m} + \frac{c_l}{12}(n^3 - n)\delta_{n+m,0}, \quad (3.60)$$

$$[\bar{L}_n, \bar{L}_m] = (n - m)\bar{L}_{n+m} + \frac{c_l}{12}(n^3 - n)\delta_{n+m,0}, \quad (3.61)$$

$$[L_n, \bar{L}_m] = 0. \quad (3.62)$$

The constant c is called the central charge and characterizes a given conformal field theory. From the above commutation relations, we have two copies of an infinite-dimensional algebra called the Virasoro algebra. A conformally invariant field theory is a representation of this algebra with a particular value of the central charge c_l . The case $c_l = 0$ corresponds to the classical symmetry algebra called the Witt algebra. The quantization of the classical systems with symmetries yields a representation of the classical symmetry group in the projective space of a Hilbert space, which is called a projective representation, i.e., a representation up to a phase. The form of algebra in (3.62) can be changed by shifting the generators $\hat{L}$ and $\hat{\bar{L}}$ by constants but this is constrained by the fact that the subalgebra of $\hat{L}_{-1}, \hat{L}_0, \hat{L}_{-1}$ satisfy

$$\begin{aligned} [L_1, L_{-1}] &= 2L_0, \\ [L_{\pm 1}, L_0] &= \pm L_{\pm 1}, \end{aligned}$$

with no central charge term. The global conformal group $SL(2, \mathbb{C})$ generated by $L_{-1,0,1}$ and $\bar{L}_{-1,0,1}$ remains an exact symmetry despite the central charge. We can easily show that the Virasoro algebra is the only possible central extension of the Witt algebra up to a trivial redefinition [13]. Consider the commutation relation of the type

$$[L_n, L_m] = (n - m)L_{n+m} + c_l f(n, m).$$

As the global symmetry $SL(2, \mathbb{C})$ does not have any central charge term we get

$$\begin{aligned} [L_1, L_{-1}] &= 2L_0 + c_l f(1, -1) = 2L_0, \\ [L_{\pm 1}, L_0] &= \pm L_{\pm 1} + c_l f(\pm 1, 0) = \pm L_{\pm 1}, \\ [L_n, L_0] &= nL_n + c_l f(n, 0) = nL_n. \end{aligned}$$

The commutation relations must obey the Jacobi identity

$$\begin{aligned} &[[L_n, L_m], L_0] + [[L_m, L_0], L_n] + [[L_0, L_n], L_m] \\ &= (n - m)[L_{n+m}, L_0] + m[L_m, L_n] - n[L_n, L_m] \\ &= (n - m)(n + m)L_{n+m} + m(m - n)L_{m+n} \\ &\quad + mc_l f(m, n) - n(n - m)L_{n+m} - nc_l f(n, m) \\ &= (n + m)c_l f(n, m) \\ &= 0. \end{aligned}$$

In the above result, we have used the fact that $f(n, m) = -f(m, n)$. From the use of the Jacobi identity, it is clear that unless $n + m = 0$, $f(n, m) = 0$. The commutation relations must also

satisfy

$$\begin{aligned}
& [[L_{-n+1}, L_n], L_{-1}] + [[L_n, L_{-1}], L_{-n+1}] + [[L_{-1}, L_{-n+1}], L_n] \\
&= (1-2n)[L_1, L_{-1}] + (n+1)[L_{n-1}, L_{-n+1}] + (n-2)[L_{-n}, L_n] \\
&= 2(1-2n)L_0 + (n+1)(2n+2)L_0 + (n+1)c_l f(n-1, 1-n) \\
&\quad - 2n(n-2)L_0 + (n-2)c_l f(-n, n) \\
&= (n+1)c_l f(n-1, 1-n) + (n-2)c_l f(-n, n) \\
&= 0.
\end{aligned}$$

This gives us a recursion relation

$$\begin{aligned}
f(n, -n) &= \frac{n+1}{n-2} f(n-1, 1-n) \\
&= \frac{(n+1)n}{(n-2)(n-3)} f(n-2, 2-n) \\
&= \frac{(n+1)!}{3!(n-2)!} f(2, -2) \\
&= \frac{(n+1)n(n-1)}{3!} \frac{1}{2} \\
&= \frac{1}{12} (n^3 - n).
\end{aligned}$$

In the above derivation, the base condition is chosen to be $f(2, -2) = \frac{1}{2}$. This is exactly the anomalous term in the commutation relation of the Virasoro algebra. Thus, Virasoro algebra is the centrally extended Witt algebra. The two-point function of the holomorphic part of the stress tensor with itself at $g = g_{*l}$ is expressed as [1]

$$\langle T(z)T(0) \rangle = \frac{c_l}{2z^4}. \quad (3.63)$$

At $g = g_{*l}$, $H = G = 0$ we obtain from (3.51)

$$c(g_{*l}) = c_l. \quad (3.64)$$

Now, we will focus on the behavior of the theory in the vicinity of the fixed point $g_{*0} \in Q$. It is convenient to choose a coordinate system in such a way that g_{*0} coincides with the origin in $Q \implies g_{*0} = 0$. We have already seen that at g_{*0} , there is a new emergent symmetry in the theory, the conformal symmetry. The space $\mathcal{A}$ for $g = 0$ contains a certain set of primary fields $\phi_\alpha \in \mathcal{A}$, which obey

$$\begin{aligned}
L_n \phi_\alpha &= \bar{L}_n \phi_\alpha = 0 \quad \text{for } n > 0 \\
L_0 \phi_\alpha &= h_\alpha \phi_\alpha, \quad \bar{L}_0 \phi_\alpha = \bar{h}_\alpha \phi_\alpha,
\end{aligned}$$

where h and $\bar{h}$ are holomorphic and anti-holomorphic conformal dimension.

$$\begin{aligned}
s_\alpha &= h - \bar{h} \\
\Delta_\alpha &= h + \bar{h}.
\end{aligned}$$

In the above expressions, s is the spin of the field and Δ coincides with the scaling dimensions of the field ϕ_α . We know that conformal symmetry fixes the form of the two-point function [1]

$$\langle \phi_\alpha(w, \bar{w}) \phi_\beta(z, \bar{z}) \rangle = \frac{C_{\alpha\beta}}{(w-z)^{2h} (\bar{w}-\bar{z})^{2\bar{h}}}. \quad (3.65)$$

Since the coefficient $C_{\alpha\beta}$ is symmetric, it is convenient to choose a normalization such that $C_{\alpha\beta} = \delta_{\alpha\beta}$. This means that the fields ϕ_α are mutually orthogonal.

$$\langle \phi_\alpha(w, \bar{w}) \phi_\beta(z, \bar{z}) \rangle = \frac{\delta_{\alpha\beta}}{(w-z)^{2h}(\bar{w}-\bar{z})^{2\bar{h}}}. \quad (3.66)$$

We can construct descendant fields spanned by all possible independent monomials of the form

$$L_{-n_1} L_{-n_2} \dots L_{-n_l} \bar{L}_{-m_1} \bar{L}_{-m_2} \dots \bar{L}_{-m_j} \phi_\alpha, \quad (3.67)$$

with $n_p, m_l > 0$. The set comprising a primary field ϕ_α and all of its descendants is called a conformal family and forms an irreducible representation of the Virasoro algebra with central charge $c_0 = c(g_{*0})$. So the (3.66) implies that orthogonality of the highest weight states is equivalent to the orthogonality of all the descendants of the two states. The operators in $\mathcal{A}$ form a closed operator algebra, and invariance under scaling transformation requires that the operator algebra have the following form

$$\phi_\alpha(z, \bar{z}) \phi_\beta(0, 0) = \sum_\gamma C_{\alpha\beta\gamma} z^{h_\gamma - h_\alpha - h_\beta} \bar{z}^{\bar{h}_\gamma - \bar{h}_\alpha - \bar{h}_\beta} (\phi_\gamma(0, 0) + \dots). \quad (3.68)$$

The terms in the round bracket in the above expression, which have been omitted, are those of higher power in $z, \bar{z}$ which describe the contribution due to descendant fields. The coefficients $C_{\alpha\beta\gamma}$ are called structure constants of the operator algebra and are symmetric in all three indices. The conformal invariance also fixes the form of the three-point function

$$\begin{aligned} \langle \phi_\alpha(z_1, \bar{z}_1) \phi_\beta(z_2, \bar{z}_2) \phi_\gamma(z_3, \bar{z}_3) \rangle &= C_{\alpha\beta\gamma} \frac{1}{z_{12}^{h_\alpha + h_\beta - h_\gamma} z_{23}^{h_\gamma + h_\beta - h_\alpha} z_{13}^{h_\alpha + h_\gamma - h_\beta}} \\ &\quad \times \frac{1}{\bar{z}_{12}^{\bar{h}_\alpha + \bar{h}_\beta - \bar{h}_\gamma} \bar{z}_{23}^{\bar{h}_\gamma + \bar{h}_\beta - \bar{h}_\alpha} \bar{z}_{13}^{\bar{h}_\alpha + \bar{h}_\gamma - \bar{h}_\beta}}. \end{aligned} \quad (3.69)$$

In the vicinity of the fixed point $g = 0$, such that $\beta(0) = 0$, we can expand $\beta^i(g)$ in Taylor series in powers of g . Let us denote $\Phi_i^0 = \Phi_i|_{g=0}$, $\Phi_i^0 \in \mathcal{A}_{g=0}^0$, for $g \neq 0$ the field Φ_i is defined by the equation (3.14). At the fixed point, the fields Φ_i^0 possess well-defined conformal dimensions, and since these are spinless fields, we have $h = \bar{h}$. The first term in the Taylor series expansion of $\beta^i(g)$ about $g = 0$ is determined by using (3.21), (3.66) and the fact that $\beta^i(0) = 0$

$$\begin{aligned} &\left[\sum_{a=i,j} \left(\frac{1}{2} x_a^\mu \frac{\partial}{\partial x_a^\mu} + \Gamma_a(0) \right) \right] \langle \Phi_i^0(z, \bar{z}) \Phi_j^0(0, 0) \rangle = 0 \\ &\Rightarrow \left[\sum_{a=i,j} \left(\frac{1}{2} x_a^\mu \frac{\partial}{\partial x_a^\mu} + \Gamma_a(0) \right) \right] \delta_{ij} (z\bar{z})^{-2h_i} = 0 \\ &\Rightarrow \left[\sum_{a=i,j} \left(\frac{1}{2} x_a^\mu \frac{\partial}{\partial x_a^\mu} + \Gamma_a(0) \right) \right] \delta_{ij} (x^2)^{-2h_i} = 0 \\ &\Rightarrow \left(-2h_i \delta_{ij} (x^2)^{-2h_i} + \langle (\Gamma \Phi_i^0)(z, \bar{z}) \Phi_j^0(0, 0) \rangle + \langle \Phi_i^0(z, \bar{z}) (\Gamma \Phi_j^0)(0, 0) \rangle \right) = 0 \\ &\Rightarrow \left(-2h_i \delta_{ij} (x^2)^{-2h_i} + \langle (\gamma_i^k \Phi_k^0)(z, \bar{z}) \Phi_j^0(0, 0) \rangle + \langle \Phi_i^0(z, \bar{z}) (\gamma_j^k \Phi_k^0)(0, 0) \rangle \right) = 0 \end{aligned}$$

The orthonormality condition tells us that the two-point correlators will vanish unless $k = j$ in the first correlator and $k = i$ in the second one. Then, for the left-hand side of the above

equation to vanish,

$$\gamma_i^j(0)\Phi_j^0 = h_i\delta_i^j\Phi_j. \quad (3.70)$$

Using (3.26) we get

$$\begin{aligned} \left(\delta_j^i - \frac{\partial\beta^i}{\partial g^j}\right)\Phi_j^0 &= h_i\delta_i^j\Phi_j^0 \\ \Rightarrow \beta^i(g) &= \epsilon_i g^i + \mathcal{O}(g^2), \end{aligned} \quad (3.71)$$

where $\epsilon_i = 1 - h_i$.

The higher-order terms in the expansion can be calculated using perturbation theory about the fixed point, but the first few terms tell us nothing about the global topological properties of the renormalization group. Here we will consider the case when the conformal dimensions h_i of the fields Φ_i^0 is close to one, i.e.,

$$1 - h_i = \epsilon_i, \quad |\epsilon_i| \sim \epsilon \ll 1. \quad (3.72)$$

When operator dimensions are near marginality, quantum corrections to their scaling behavior become important. This is the logic behind the ϵ -expansion or near-marginal analysis of the fixed points. In this case, the non-linear terms become comparable to the linear part when $g^i \sim \epsilon$, then the renormalization group possesses non-trivial behavior, like having other fixed points, in the region $g^i \lesssim \epsilon$. This can be understood simply if we consider the one-charge case $n = 1$. In this case the Taylor series expansion of the $\beta(g)$ takes the form

$$\beta(g) = \epsilon_1 g + c_2 g^2 + \mathcal{O}(g^3).$$

If $\epsilon \ll 1$, then the coefficient of the linear term is small. The one-loop correction $c_2 g^2$ become important and comparable to the linear term when

$$\epsilon g \sim c_2 g^2 \implies g \sim \frac{\epsilon}{c_2} \equiv \mathcal{O}(\epsilon).$$

In this case, the higher-order terms become comparable to the linear term, and we can no longer truncate $\beta(g)$ at linear order. Now, requiring that $\beta(g_*) = 0$ gives us a non-trivial fixed point

$$g \approx \frac{\epsilon}{c_2}.$$

Thus the renormalization group exhibits non-trivial behavior in the region $g \lesssim \epsilon$. This means we can safely use perturbation theory around the fixed point, as the perturbation theory around the fixed point amounts to a Taylor expansion of the functions about $g = 0$. The Taylor series will converge only if there exists a non-trivial fixed point in the vicinity of $g = 0$. Past this point, the series diverges, and the perturbation theory breaks down. To calculate the next term in the expansion (3.71) in case (3.72) we use (3.15)

$$\begin{aligned} \frac{\partial}{\partial g^k} \langle \Phi_i(x) \Phi_j(0) \rangle|_{g=0} &= \\ \langle (B_k^0 \Phi_i^0)(x) \Phi_j^0(0) \rangle + \langle \Phi_i^0(x) (B_k^0 \Phi_j^0)(0) \rangle + \int d^2 y \langle \Phi_i^0(x) \Phi_j^0(0) \Phi_k^0(y) \rangle, \end{aligned} \quad (3.73)$$

where $B_k^0 = B_k|_{g=0}$. Let us first evaluate the integral on the right-hand side of (3.73) using (3.69)

$$\begin{aligned} \int d^2y \langle \Phi_i^0(x) \Phi_j^0(0) \Phi_k^0(y) \rangle &= \int d^2y \frac{C_{ijk}}{(x^2)^{h_i+h_j-h_k} |\mathbf{x}-\mathbf{y}|^{2(h_i+h_k-h_j)} (y^2)^{h_j+h_k-h_i}} \\ &\quad + \langle (b_k^0 \Phi_i^0)(x) \Phi_j^0(0) \rangle + \langle \Phi_i^0(x) (b_k^0 \Phi_j^0)(0) \rangle. \end{aligned}$$

The terms with the operator b_k appear in case of divergence of the integral. The exact form of b_k depends on the cutoff method and will not be required for further calculation, so we will ignore it. Now, let us focus on the first term in the above expression

$$\begin{aligned} \int d^2y \langle \Phi_i^0(x) \Phi_j^0(0) \Phi_k^0(y) \rangle &= \int d^2y \frac{C_{ijk}}{(x^2)^{h_i+h_j-h_k} |\mathbf{x}-\mathbf{y}|^{2(h_i+h_k-h_j)} (y^2)^{h_j+h_k-h_i}} \\ &= C_{ijk} (x^2)^{h_k-h_i-h_j} \int d^2y \frac{1}{|\mathbf{x}-\mathbf{y}|^{2(h_i+h_k-h_j)} (y^2)^{h_j+h_k-h_i}}. \end{aligned}$$

The above integral is a convolution integral and can be evaluated by the fact that the inverse Fourier transform of a product of Fourier transforms is the convolution of the original functions, i.e., consider two functions $f(x)$ and $g(x)$ with Fourier transforms $F(k)$ and $G(k)$ then the convolution of these functions is defined as

$$\begin{aligned} (f * g)(x) &= \int_{-\infty}^{\infty} dy g(y) f(x-y) \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} g(y) \int_{-\infty}^{\infty} F(k) \exp(ik(x-y)) dk dy \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} F(k) \left[\int_{-\infty}^{\infty} g(y) \exp(-iky) dy \right] \exp(ikx) dk \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} F(k) G(k) \exp(ikx) dk. \end{aligned}$$

The above result is easily generalized to higher dimensions. The Fourier transform of $(y^2)^{-h_j-h_k+h_i}$ is

$$\begin{aligned} &= \int (y^2)^{-h_j-h_k+h_i} \exp(-i\mathbf{k} \cdot \mathbf{y}) d^2y \\ &= \int_0^{\infty} dy y^{1-2(h_j+h_k-h_i)} \int_0^{2\pi} \exp(-iky \cos \theta) d\theta \\ &= 2\pi \int_0^{\infty} y^{1-2(h_j+h_k-h_i)} J_0(ky) dy, \end{aligned}$$

where $J_0(ky)$ is the Bessel function of the first kind. Now, substitute $ky = t$

$$\begin{aligned} &= (2\pi) k^{2(h_j+h_k-h_i)-2} \int_0^{\infty} t^{1-2(h_j+h_k-h_i)} J_0(t) dt \\ &= (2\pi) k^{2(h_j+h_k-h_i)-2} 2^{1-2(h_j+h_k-h_i)} \frac{\Gamma(1+h_i-h_j-h_k)}{\Gamma(h_j+h_k-h_i)}. \end{aligned}$$

Now, using the result of the Convolution theorem,

$$\begin{aligned}
C_{ijk}(x^2)^{h_k-h_i-h_j} \int d^2y \frac{1}{|\mathbf{x}-\mathbf{y}|^{2(h_i+h_k-h_j)} (y^2)^{h_j+h_k-h_i}} \\
= C_{ijk}(x^2)^{h_k-h_i-h_j} \frac{1}{(2\pi)^2} 2^{2-4h_k} \\
\times \frac{\Gamma(1+h_i-h_j-h_k)\Gamma(1+h_j-h_i-h_k)}{\Gamma(h_j+h_k-h_i)\Gamma(h_i+h_k-h_j)} \int d^2k k^{4h_k-4} \exp(i\mathbf{k} \cdot \mathbf{x}) \\
C_{ijk}(x^2)^{h_k-h_i-h_j} \int d^2y \frac{1}{|\mathbf{x}-\mathbf{y}|^{2(h_i+h_k-h_j)} (y^2)^{h_j+h_k-h_i}} \\
= C_{ijk}(x^2)^{h_k-h_i-h_j} 2^{2-4h_k} \frac{\Gamma(1+h_i-h_j-h_k)\Gamma(1+h_j-h_i-h_k)}{\Gamma(h_j+h_k-h_i)\Gamma(h_i+h_k-h_j)} \\
\times \int_0^\infty dk k^{4h_k-4} \int_0^{2\pi} \exp(ikx \cos \theta) d\theta \\
= (2\pi) C_{ijk}(x^2)^{h_k-h_i-h_j} 2^{2-4h_k} \frac{\Gamma(1+h_i-h_j-h_k)\Gamma(1+h_j-h_i-h_k)}{\Gamma(h_j+h_k-h_i)\Gamma(h_i+h_k-h_j)} \\
\times \int_0^\infty dk k^{4h_k-4} J_0(kx).
\end{aligned}$$

This gives us

$$= (x^2)^{1-h_i-h_j-h_k} C_{ijk} I_{ij}^k, \quad (3.74)$$

where

$$I_{ij}^k = (2\pi) \frac{\Gamma(1+h_i-h_j-h_k)\Gamma(1+h_j-h_i-h_k)\Gamma(2h_k-1)}{\Gamma(h_j+h_k-h_i)\Gamma(h_i+h_k-h_j)\Gamma(2-2h_k)}. \quad (3.75)$$

Now, using (3.72) above equation can be rewritten as

$$I_{ij}^k = (2\pi) \frac{\Gamma(\epsilon_j + \epsilon_k - \epsilon_i)\Gamma(\epsilon_i + \epsilon_k - \epsilon_j)\Gamma(1-2\epsilon_k)}{\Gamma(1-\epsilon_i - \epsilon_k + \epsilon_j)\Gamma(1-\epsilon_j - \epsilon_k + \epsilon_i)\Gamma(2\epsilon_k)}. \quad (3.76)$$

To simplify it further, we make the approximation

$$\Gamma(\epsilon) = \frac{1}{\epsilon} - \gamma + \mathcal{O}(\epsilon^2), \quad (\gamma \text{ is the Euler-Mascheroni constant}), \quad (3.77)$$

and

$$\Gamma(1-\epsilon) \approx 1 + \epsilon\gamma + \mathcal{O}(\epsilon^2). \quad (3.78)$$

Using these approximations (3.76) becomes

$$I_{ij}^k = \frac{4\pi\epsilon_k}{(\epsilon_i + \epsilon_k - \epsilon_j)(\epsilon_j + \epsilon_k - \epsilon_i)} (1 + \mathcal{O}(\epsilon^2)). \quad (3.79)$$

We have already showed that at $\tau = 0 \implies x^2 = 1$ the function G depends only on the couplings, so it is convenient to choose a renormalization condition such that (3.73) vanishes at $x^2 = 1$. This corresponds to a special choice of coordinate system in Q

$$G_{ij}(g) = \delta_{ij} + \mathcal{O}(g^2). \quad (3.80)$$

This is the Taylor expansion of the function G around $g = 0$. Now requiring that (3.73) vanishes at $x^2 = 1$ we get

$$\langle (B_k^0 \Phi_i^0)(x) \Phi_j^0(0) \rangle|_{x^2=1} + \langle \Phi_i^0(x) (B_k^0 \Phi_j^0)(0) \rangle|_{x^2=1} = -C_{ijk} I_{ij}^k. \quad (3.81)$$

Consider $\langle (B_k^0 \Phi_i^0)(x) \Phi_j^0(0) \rangle|_{x^2=1}$, this will be non-zero only when the action of B_k^0 on Φ_i^0 will project onto Φ_j^0 as we have chosen the normalization condition in such a way (3.66).

So we expect $B_k^0 \Phi_i^0 = \sum_l \alpha_{il}^{(k)} \Phi_l + (\text{other terms})$. "Other terms" means there could be terms, like the descendant states, which will vanish in the correlator and will not contribute. Since in our case the normalization is chosen such that the structure constants of the operator product expansion (3.68) and the three-point function (3.69) are similar. Thus, $\alpha_{ij}^k \propto C_{ijk}$, so we redefine $\alpha_{ij}^k \rightarrow C_{ijk} \alpha_{ij}^k$. This gives us

$$\alpha_{ij}^k + \alpha_{ji}^k = -I_{ij}^k. \quad (3.82)$$

The simplest solution to the above equation is to define

$$\mathcal{I}_{ik}^j = \frac{1}{2} (I_{ij}^k + I_{jk}^i - I_{ik}^j) = \frac{2\pi}{\epsilon_i + \epsilon_k - \epsilon_j} (1 + \mathcal{O}(\epsilon^2)). \quad (3.83)$$

Using the above relation we have $\mathcal{I}_{ik}^j + \mathcal{I}_{jk}^i = I_{ij}^k$. So we have $\alpha_{ij}^k = -\mathcal{I}_{ik}^j$ and $\alpha_{ji}^k = -\mathcal{I}_{jk}^i$. Using the fact that the form of the two-point correlator has the form (3.66), we get

$$\begin{aligned} & \frac{\partial}{\partial g^k} \langle \Phi_i(x) \Phi_0(0) \rangle|_{g=0} \\ &= (x^2)^{-h_i-h_j} C_{ijk} \left[\mathcal{I}_{ik}^j \{ (x^2)^{1-h_k} - (x^2)^{h_i-h_j} \} + \mathcal{I}_{jk}^i \{ (x^2)^{1-h_k} - (x^2)^{h_j-h_i} \} \right]. \end{aligned} \quad (3.84)$$

We have already computed the leading order term in the Taylor expansion of the matrix of anomalous dimension in (3.70). To compute the next-to-leading order term, we have to multiply the above result by β^k , sum over k , and compare it with the Callan-Symanzik equation (3.21). To do so, we first write the Taylor series expansion of the matrix of anomalous dimension as

$$\gamma_i^j(g) = h_i \delta_i^j + \sum_k g^k \alpha_{ki}^j + \mathcal{O}(g^2). \quad (3.85)$$

The leading order term in the matrix of anomalous dimension will cancel out with the derivative of $\langle \Phi_i(x) \Phi_j(0) \rangle|_{g=0}$. Then comparing the coefficients of powers of x^2 gives us

$$\begin{aligned} \alpha_{ki}^j &= (\epsilon_i + \epsilon_k - \epsilon_j) C_{ijk} \mathcal{I}_{ik}^j \\ &= 2\pi C_{ijk} + \mathcal{O}(\epsilon^2). \end{aligned}$$

Now using (3.26) we get

$$\begin{aligned} \left(\delta_i^j - \frac{\partial \beta^j}{\partial g^i} \right) &= h_i \delta_i^j + C_{ki}^j g^k \\ \Rightarrow \frac{\partial \beta^j}{\partial g^i} &= \epsilon_i \delta_i^j - C_{ki}^j g^k \end{aligned}$$

This implies that

$$\beta^j(g) = \epsilon_j g^j - \frac{1}{2} C_{ki}^j g^k g^i + \mathcal{O}(g^3), \quad (3.86)$$

where $C_{ki}^j = (\epsilon_i + \epsilon_k - \epsilon_j)C_{ijk}\mathcal{T}_{ik}^j = 2\pi C_{ijk} + \mathcal{O}(\epsilon^2)$. (3.56) tells us that the gradient of the c -function is proportional to the β -function. Using (3.86) we can compute $c(g)$ to be

$$c(g) = c_0 - 6\epsilon_i g^i g^i + 2C_{ijk} g^i g^j g^k + \mathcal{O}(g^3). \quad (3.87)$$

The constant term in the above expansion is determined from the fact that at the fixed point $g_{*l} = 0$ in our choice of the coordinates, the function $c(g)$ coincides with the central charge of the conformal field theory. The metric G_{ij} in (3.56) can be used to raise and lower the indices. Using this fact and the expansion of the $c(g)$ -function, it is easily verified that up to the given order, the following relation holds

$$\beta^i(g) = -\frac{1}{12}G^{ij}(g)\frac{\partial c(g)}{\partial g^j}. \quad (3.88)$$

As the metric $G_{ij}(g)$ is positive definite, the above relation tells us that at fixed points $\beta^i(g_*) = 0$ the function $c(g)$ is stationary, meaning

$$\beta_i(g) = 0 \implies \frac{\partial c(g)}{\partial g^i} = 0. \quad (3.89)$$

This proves the second property about the structure of the renormalization group stated in 3.1. We have shown that at the fixed point, the function (3.51) is stationary, which implies that the two-point function $G(\tau)$ vanishes at the fixed point, which using positivity condition implies that the $T_\mu^\mu = T_{zz} = 0$, the trace vanishes and we have full conformal symmetry. This shows that in two dimensions, scale invariance implies conformal invariance. (3.56) tell us that the renormalization group flux described by (3.5) leads to a decrease of $c(g)$. The function $c(g)$ has an interpretation of counting the number of degrees of freedom of the system. Then (3.52) tells us that as we flow from UV to IR, the number of degrees of freedom continuously decreases. Physics changes monotonically as we look at longer and longer scales. This monotonicity implies that at long distances the theory will approach a stationary point of $c(g)$ and hence a conformal field theory. This can be understood intuitively that at long distances the theory should forget about the underlying scales. The perturbation theory about the fixed point showed the properties of the renormalization group structure in the vicinity of the fixed point.

Chapter 4

Does Scale Invariance imply Conformal Invariance?

4.1 Introduction

In the previous chapter, we showed the proof due to Zamolodchikov [17] of the enhancement of scale invariance to conformal invariance in two dimensions. In this chapter, we will derive the conserved scale and conformal current, the form of which was argued in chapter 2. We will then show the conditions imposed by the conservation of these currents on the trace of the stress tensor, using which we will be able to construct an improved stress tensor with vanishing trace. We will then explore some examples of the theories that are scale invariant but not necessarily conformally invariant. The proof of enhancement of scale invariance to conformal invariance in the previous chapter was based on the assumption that the stress tensor scales according to its canonical dimension, and we did not take into account the effect of new terms that can be introduced due to the renormalization group flow. Here we will show that under the broad assumptions of unitarity and discreteness of the scaling dimension that we can construct a new improved stress tensor which scales according to the canonical scaling dimension. This will be an extension to the argument due to Zamolodchikov.

4.2 The Construction of a Traceless Energy-Momentum Tensor

We have demonstrated in chapter 2, from the variation of the action, that the trace of the stress tensor vanishes in the case of conformal invariance. In general, the canonically obtained stress tensor using Noether's theorem may not have a vanishing trace. In this section, we will demonstrate the conditions for the existence of a conserved scale and a conformal current and how we can rewrite the stress tensor so that the trace vanishes in the case of conformal invariance. We begin with a Poincaré invariant field theory defined on a flat d -dimensional spacetime, allowing for either Euclidean or Minkowski signature. We will focus on two infinitesimal transformations

1. scale transformations $x^\mu \rightarrow \lambda x^\mu$,
2. conformal transformations $\delta x^\mu = \epsilon v^\mu(x)$ such that

$$\partial_\mu v_\nu(x) + \partial_\nu v_\mu(x) = \frac{2}{d} \eta_{\mu\nu} \partial \cdot v(x) \quad (4.1)$$

We have demonstrated the derivation of the above equation in chapter 2. Following the derivation in [3] we arrive at

$$(2 - d) \partial_\mu \partial_\nu (\partial \cdot v(x)) = \eta_{\mu\nu} \partial^2 (\partial \cdot v(x)) \quad (4.2)$$

and contracting it with $\eta^{\mu\nu}$ we get

$$(d-1)\partial^2(\partial \cdot v(x)) = 0. \quad (4.3)$$

(4.1) implies that $v^\mu(x)$ are local change of scales which preserve angles. As we have already seen in chapter 2, conformal transformations include Poincaré, scale and special conformal transformation, and due to vanishing of left-hand side of (4.2) due to (4.3) we see that $\partial \cdot v(x)$ spans all linear functions of x^μ . In $d = 2$ as we saw in chapter 3, the symmetry group is infinite dimensional and from (4.3) it follows that $\partial \cdot v(x)$ spans all harmonic functions of x^μ .

Now we will derive the form of the scale current as discussed in chapter 2 using the methods described in [15]. If the theory under consideration is scale invariant, then we know from Noether's argument that any conserved current gives rise to time-independent charge [11]

$$Q \equiv \int_\sigma d\sigma_\mu J^\mu(x). \quad (4.4)$$

These charges corresponding to different currents satisfy the underlying group algebra we studied in chapter 2. We also have the conservation law for the theory to be invariant under the transformations generated by Q

$$\partial_\mu J^\mu(x) = 0. \quad (4.5)$$

Because of the above condition, we can deform σ without changing the value of Q , which implies that $dQ/dt = 0$. We are thus free to choose the surface $t = \text{constant}$.

$$Q = \int_{t=\text{const}} d^3x J^0(x). \quad (4.6)$$

The time dependence of the operators in the Heisenberg picture is governed by the equation

$$\frac{dQ}{dt} = i[H, Q] + \frac{\partial Q}{\partial t}, \quad (4.7)$$

where $H \equiv P^0 \equiv \int_{t=\text{const}} d^3x T^{00}(x)$. In chapter 2, we have seen the commutation relations for the conformal group, and the conservation law $dQ/dt = 0$ is the consequence of the continuity equation satisfied by the conserved current. Now, in the above equation, if we substitute $Q = D$, where D is generator of scale transformation, we get

$$i[H, D] + \frac{\partial D}{\partial t} = 0. \quad (4.8)$$

From chapter 2 we know $[P^\mu, D] = -iP^\mu$, using this

$$P^0 + \frac{\partial}{\partial t} \int_{t=\text{const}} d^3x S^0(x) = 0, \quad (4.9)$$

where $S^\mu(x)$ is the scale current. For the above equation to hold, the part of D which depends on t must be of form

$$- \int_{t=\text{const}} d^3x T^{00}(x)t. \quad (4.10)$$

This means we have the integrated relation apart from the terms that do not contribute to the surface integral

$$S^0 = T^{00}(x)t + K^0(x). \quad (4.11)$$

This is the zeroth component of the scale current. Lorentz invariance requires that this be of the form

$$S^\mu(x) = T^{\mu\nu}(x)x_\nu + K^\mu(x), \quad (4.12)$$

where $K^\mu(x)$ has coordinate dependence only through fields. This is because if there is any additional explicit coordinate dependence in K^μ , this would violate the conformal algebra, as can be seen from (4.11) as

$$i[P^0, D] = -\frac{\partial}{\partial t} \int_{t=\text{const}} d^3x (T^{00}(x)t + K^0(x)) = P^0. \quad (4.13)$$

If $K^\mu(x)$ had explicit coordinate dependence, then we would have additional terms which would violate the commutation relation, and therefore $K^\mu(x)$ can have no explicit coordinate dependence. Conservation of (4.12) implies that

$$T^\mu_\mu(x) = -\partial_\mu K^\mu(x). \quad (4.14)$$

There is an ambiguity in the definition of $S^\mu(x)$. We know that, in general, many symmetric conserved stress tensors give rise to the same Hamiltonian. This means we can modify our stress tensor by adding terms that respect the symmetry of the stress tensor and still satisfy the conservation equation. If $T_{\mu\nu}(x)$ and $T'_{\mu\nu}(x)$ are such tensors their difference must be of form [12]

$$T'_{\mu\nu}(x) - T_{\mu\nu}(x) = \partial^\sigma \partial^\rho Y_{\mu\sigma\nu\rho}(x), \quad (4.15)$$

for some local operator $Y_{\mu\sigma\nu\rho}(x)$. This operator is anti-symmetric on $\mu\sigma$ and $\nu\rho$, and symmetric under $\mu\sigma$ with $\nu\rho$. If $T_{\mu\nu}(x)$ satisfies (4.14) then so does $T'_{\mu\nu}(x)$ with $K^\mu(x)$ replaced by $K^\mu - \partial^\rho Y^{\mu\nu}_{\rho\nu}$. This implies that (4.12) is a necessary and sufficient condition for the existence of a conserved scale current, i.e., the trace of stress is equal to the divergence of a local operator.

We proceed in a similar way to derive the conformal current

$$i[P^0, K^\mu] + \frac{\partial K^\mu}{\partial t} = 0. \quad (4.16)$$

This can be written using the commutation relation from chapter 2 as

$$2(\eta^{\mu 0} D - J^{\mu 0}) = -\frac{\partial}{\partial t} \int_{t=\text{const}} d^3x A^{\mu 0}(x), \quad (4.17)$$

where $A^{\mu\nu}(x)$ is the conformal current. We can write $J^{\mu 0} = \int_{t=\text{const}} (T^{00}(x)x^\mu - T^{\mu 0}x^0)$, this gives us up to an overall term which does not contribute to the surface integral

$$A^{\mu 0} = -2K^\mu t + 2T^{00}x^\mu t - T^{0\mu}t^2 + L^{\mu 0}(x). \quad (4.18)$$

Lorentz invariance requires that this be of the form

$$\begin{aligned} A^{\mu\nu}(x) &= -2K^\nu x^\mu - 2T^{\nu\rho}(x)x^\mu x_\rho + T^{\nu\mu}(x)x^2 + L^{\mu\nu}(x) \\ &= (\eta^{\nu\rho}x^2 - 2x^\nu x^\rho)T^\mu_\rho(x) - 2x^\nu K^\mu(x) + L^{\mu\nu}(x) \end{aligned} \quad (4.19)$$

The finite conformal transformation is given as (2.25), if we write $b^\mu = \epsilon c^\mu$ where $\epsilon \ll 1$ then the infinitesimal version is given as

$$\delta x^\mu = \epsilon \{2(c \cdot x)x^\mu - c^\mu x^2\}. \quad (4.20)$$

Now comparing this with $\delta x^\mu = b^\rho \delta_\rho x^\mu$ we get

$$v^\mu(x) = c^\rho \delta_\rho x^\mu = c^\rho (2x_\rho x^\mu - \delta_\rho^\mu x^2) \quad (4.21)$$

This can be written as

$$v^\mu(x) = -c^\rho g_{\rho\nu} (g^{\nu\mu} x^2 - 2x^\nu x^\mu) = -c_\nu (g^{\nu\mu} x^2 - 2x^\nu x^\mu). \quad (4.22)$$

Now we compute $\partial \cdot v(x)$, this is just

$$\partial \cdot v(x) = 2c^\rho x_\rho, \quad (4.23)$$

and $\partial_\nu (\partial \cdot v(x)) = 2c_\nu$. From Noether's theorem [3] we know that $j^\mu = -T^\mu_\nu(x) \delta x^\nu$. If we contract (4.19) with c_ν and make the above substitutions we get

$$j^\mu_v(x) = v^\nu(x) T^\mu_\nu(x) + (\partial \cdot v(x)) K^\mu(x) + \partial_\nu (\partial \cdot v(x)) L^{\mu\nu}(x). \quad (4.24)$$

The first term determines the spacetime nature of the conformal transformation, the second term appears because the conformal transformation is a local change of scale, and the third term appears because of the position dependence of the scale factor. The current (4.24) must be conserved, in $d \geq 3$ we showed that $\partial \cdot v(x)$ is a linear function of coordinates, so the conservation of (4.24) gives

$$\partial_\mu v^\nu(x) T^\mu_\nu(x) + \partial_\mu (\partial \cdot v(x)) K^\mu(x) + (\partial \cdot v(x)) \partial_\mu K^\mu(x) + \partial_\nu (\partial \cdot v(x)) \partial_\mu L^{\mu\nu}(x) = 0. \quad (4.25)$$

Now, comparing the coefficients tells us that the conservation of (4.24) implies that

$$T^\mu_\mu(x) = -\partial_\mu K^\mu(x) \text{ and } K^\mu(x) = -\partial_\nu L^{\mu\nu}(x). \quad (4.26)$$

In $d = 2$, $\partial \cdot v(x)$ is a general harmonic function of coordinates, and conservation of (4.24) implies

$$\begin{aligned} \partial_\mu v^\nu(x) T^\mu_\nu(x) + \partial_\mu (\partial \cdot v(x)) K^\mu(x) + (\partial \cdot v(x)) \partial_\mu K^\mu(x) + \partial_\nu (\partial \cdot v(x)) \partial_\mu L^{\mu\nu}(x) \\ + \partial_\mu \partial_\nu (\partial \cdot v(x)) L^{\mu\nu}(x) = 0. \end{aligned} \quad (4.27)$$

This gives an additional relation for conservation $L^{\mu\nu}(x) = g^{\mu\nu} L(x)$ for some local operator $L(x)$, for the last term in the above equation to vanish independently. Thus, we have the following conditions

$$T^\mu_\mu(x) = \partial_\mu \partial_\nu L^{\mu\nu}(x), \text{ for } d \geq 3, \quad (4.28)$$

$$T^\mu_\mu(x) = \partial^2 L(x), \text{ for } d = 2. \quad (4.29)$$

This is the necessary and sufficient condition for the existence of a conserved conformal current. If condition (4.29) holds, then we can construct an equivalent stress tensor [12]

$$\begin{aligned} \Theta_{\mu\nu}(x) = T_{\mu\nu}(x) + \frac{1}{D-2} \left(\partial_\mu \partial_\alpha L^\alpha_\nu(x) + \partial_\nu \partial_\alpha L^\alpha_\mu(x) - \partial^2 L_{\mu\nu}(x) - g_{\mu\nu} \partial_\alpha \partial_\beta L^{\alpha\beta}(x) \right) \\ + \frac{1}{(D-2)(D-1)} (g_{\mu\nu} \partial^2 L^\alpha_\alpha(x) - \partial_\mu \partial_\nu L^\alpha_\alpha(x)) \quad \text{in } D \geq 3, \end{aligned} \quad (4.30)$$

or, equivalently, in two dimensions

$$\Theta_{\mu\nu}(x) = T_{\mu\nu}(x) + \frac{1}{D-1} (\partial_\mu \partial_\nu L(x) - g_{\mu\nu} \partial^2 L(x)) \quad \text{in } D = 2. \quad (4.31)$$

It is easy to verify from the above equations that

$$\Theta^\mu_\mu(x) = 0. \quad (4.32)$$

This implies that a system will be scale invariant and not conformally invariant if the trace of the stress tensor is the divergence of the local operator $-K^\mu$, which is not itself a conserved current plus a divergence in $d \geq 3$ or a gradient in $d = 2$. This leads to the conclusion that scale invariance will imply conformal invariance if there is no local operator $K^\mu(x)$ such that the trace of the stress tensor can be written as the divergence of this local operator. In the next section, we will look at some examples of the scale-invariant field theory that may or may not be conformally invariant.

4.3 Scale vs. Conformal Invariance: Examples

In this section, we will show some examples of the theories that are scale invariant but may or may not be conformally invariant.

The first example that we will consider is the free massless scalar theory, which has the action

$$S = \frac{1}{2} \int d^d x \partial^\mu \phi \partial_\mu \phi. \quad (4.33)$$

Under scale transformation $x \rightarrow \lambda x$ the fields transform as $\phi \rightarrow \lambda^{-\Delta} \phi$, where Δ is the scaling dimension of the scalar field. Then the action is scale invariant if $\Delta = \frac{d-2}{2}$. The stress tensor can be computed using the Noether's prescription and is given by

$$T^{\mu\nu} = \partial^\mu \phi \partial^\nu \phi - \frac{g^{\mu\nu}}{2} (\partial^\alpha \phi)(\partial_\alpha \phi). \quad (4.34)$$

The trace is

$$T^\mu_\mu = \frac{(2-d)}{2} (\partial^\alpha \phi)(\partial_\alpha \phi). \quad (4.35)$$

It is clear from the above equation that the massless free scalar field theory is conformally invariant in $d = 2$. We can construct the conserved scale current according to (4.12) with

$$K^\mu = \frac{d-2}{2} \phi \partial^\mu \phi. \quad (4.36)$$

We can further show that $K^\mu = \partial_\nu L^{\nu\mu}$ such that

$$L^{\mu\nu} = \frac{d-2}{2} g^{\mu\nu} \phi^2, \quad (4.37)$$

implying that the massless scalar free field theory is conformally invariant in any dimension. If we consider the interacting ϕ^4 theory with the action

$$S = \int d^d x \left(\frac{1}{2} (\partial^\mu \phi)^2 - \frac{\Lambda}{4} \phi^4 \right). \quad (4.38)$$

This theory is scale invariant only in $d = 4$ as the coupling constant Λ is dimensionless in $d = 4$. The stress tensor is

$$T^{\mu\nu} = \partial^\mu \phi \partial^\nu \phi - \frac{g^{\mu\nu}}{2} (\partial^\alpha \phi)^2 + \frac{g^{\mu\nu}}{4} \Lambda \phi^4, \quad (4.39)$$

and the trace is

$$T^\mu_\mu = -(\partial_\mu \phi)^2 + \Lambda \phi^4. \quad (4.40)$$

In this case as well, we can construct scale current (4.12) with

$$K^\mu = \phi \partial^\mu \phi, \quad (4.41)$$

which satisfies the (4.14) on-shell. This theory is conformally invariant in $d = 4$ as we can write $K^\mu = \partial_\nu L^{\nu\mu}$ and the trace vanishes. Using (4.30) we can construct the traceless stress tensor

$$T^{\mu\nu} = \partial^\mu \phi \partial^\nu \phi - \frac{g^{\mu\nu}}{2} (\partial^\alpha \phi)^2 + \frac{g^{\mu\nu}}{4} \phi^4 - \frac{1}{6} (g^{\mu\nu} \partial^2 - \partial^\mu \partial^\nu) \phi^2. \quad (4.42)$$

In $d = 4$, due to quantum corrections, the one-loop β function is nonzero, so at the quantum level, the scale invariance is broken unless $\Lambda = 0$, the free theory, or at a non-trivial fixed point like the Wilson-Fischer fixed point in $4 - \epsilon$ dimensions [11].

Let us consider the free massless Dirac fermion, which has the action

$$S = \frac{1}{2} \int d^d x (\bar{\psi} \gamma^\mu \partial_\mu \psi - \partial_\mu \bar{\psi} \gamma^\mu \psi). \quad (4.43)$$

The stress tensor can be computed to be

$$T^{\mu\nu} = i \frac{1}{2} \bar{\psi} (\gamma^\mu \partial^\nu + \gamma^\nu \partial^\mu) \psi - i g^{\mu\nu} \bar{\psi} \gamma^\rho \partial_\rho \psi. \quad (4.44)$$

Using the Dirac equation, it is easily seen that the trace vanishes. This implies that the free massless Dirac fermion is conformally invariant in any dimension.

Next, we will consider the free Maxwell theory, the action is

$$S = \frac{1}{4} \int d^d x \sqrt{-g} F^{\mu\nu} F_{\mu\nu}. \quad (4.45)$$

The stress tensor can be computed as the functional derivative of the action with respect to the metric evaluated in flat space, which gives

$$T^{\mu\nu} = F^{\mu\rho} F_\rho^\nu - \frac{n^{\mu\nu}}{4} F^{\alpha\beta} F_{\alpha\beta}. \quad (4.46)$$

The trace is given as

$$T^\mu_\mu = \frac{4-d}{4} F^{\alpha\beta} F_{\alpha\beta} = \frac{4-d}{8} \partial_\alpha (A_\beta F^{\alpha\beta}). \quad (4.47)$$

The second equality follows from the anti-symmetric properties of $F^{\mu\nu}$. It is clear from the above equation that the trace vanishes for $d = 4$. It can be seen from the above equation that the trace can be written as the divergence of the local operator

$$K^\mu = \frac{4-d}{8} A_\nu F^{\mu\nu}. \quad (4.48)$$

But this cannot be written as the divergence of some other local operator $L^{\mu\nu}$, so the $U(1)$ Maxwell theory is scale invariant for all dimensions, but conformally invariant only in $d = 4$. The invariance under conformal transformation for relativistic systems was first discussed for the case of Maxwell theory at the beginning of 20th century. This observation is essentially due to the fact that the spacetime is four-dimensional.

These examples suggest that there is no clear relation between scale and conformal invariance. The first example of the massless scalar field theory tells us that for $d \neq 2$ the theory is conformally invariant, but the canonical stress tensor is not traceless and requires an improvement term according to (4.30). In case of massless interacting ϕ^4 theory, it is classically scale and conformally invariant only in $d = 4$, but at the quantum level, the scale invariance is broken. The massless free Dirac fermion is both scale and conformally invariant

in all dimensions. The $U(1)$ Maxwell theory is only conformally invariant in $d = 4$, but is scale invariant in all dimensions. These examples suggest that although scale invariance is required for conformal invariance, it is not a sufficient condition.

4.4 Zamolodchikov's c-theorem

We have already demonstrated the c-theorem in chapter 3, but there we assumed that the stress tensor has canonical scaling dimensions. But this is not generally true, the canonical scaling dimension of the stress tensor is violated when it is not an eigenoperator under dilation. This means that in the proof presented in the previous chapter, we did not take into account the additional terms that can be added to the stress tensor as demonstrated in 4.2, which still obey the symmetries and conservation of the stress tensor. In [12], Polchinski shows that this is still consistent and completes Zamolodchikov's argument in [17] under broader conditions.

Consider the Euclidean two-point functions of the stress tensor in complex coordinates

$$\begin{aligned} C(x^2) &= 2z^4 \langle T_{zz}(x) T_{zz}(0) \rangle, \\ E(x^2) &= x^2 z^2 \langle T_{zz}(x) T_{z\bar{z}}(0) \rangle, \\ F(x^2) &= x^4 \langle T_{z\bar{z}}(x) T_{z\bar{z}}(0) \rangle. \end{aligned} \quad (4.49)$$

As we demonstrated in 3.3, using conservation of $T_{\mu\nu}$ we obtain

$$F(x^2) = -\frac{1}{12} x^2 \frac{\partial}{\partial x^2} \{ C(x^2) - 4E(x^2) - 6F(x^2) \}. \quad (4.50)$$

If at a fixed point, the stress tensor scales according to its canonical dimension, meaning the commutator of the scale generator D with the stress tensor takes its canonical form, which in d -dimensions is

$$i[D, T_{\mu\nu}(x)] = x^\alpha \partial_\alpha T_{\mu\nu}(x) + d T_{\mu\nu}(x), \quad (4.51)$$

and if $\hat{S} = 0$ and $S|0\rangle = 0$, this means that scale symmetry is an exact symmetry as $[S, H] = 0$ and the vacuum is invariant meaning under renormalization group flow the scale symmetry is unbroken, then any correlator must be invariant under infinitesimal dilations and so $C(x^2)$, $E(x^2)$ and $F(x^2)$ will be independent of x^2 . This implies that the right-hand side of (4.50) vanishes and the two point correlator of $T_\mu^\mu = T_{z\bar{z}}$ is zero, and reflection positivity as discussed in 2 implies $T_\mu^\mu = 0$ as an operator equation. This is the result that we proved in 3 to show that in $d = 2$ scale invariance is enhanced to conformal invariance.

This argument is not complete as it assumes the canonical scaling dimension of the stress tensor, which is not correct, as there can be improvement terms in the stress tensor due to the renormalization group. The general commutator of the scale generator with the stress tensor is given as

$$i[D, T_{\mu\nu}(x)] = x^\alpha \partial_\alpha T_{\mu\nu}(x) + d T_{\mu\nu}(x) + \partial^\sigma \partial^\rho \hat{Y}_{\mu\sigma\nu\rho}(x), \quad (4.52)$$

where $\hat{Y}_{\mu\sigma\nu\rho}(x)$ has same symmetries as $Y_{\mu\sigma\nu\rho}$ in (4.15). The above commutator simply states the fact that the stress tensor is not an eigenoperator under dilation. We have the freedom to redefine $T_{\mu\nu}$ 4.2, we can use this to bring the commutator back to its canonical form. This can be done as follows [12], let $Y_{\mu\sigma\nu\rho}^a(x)$ be a complete set of operators of appropriate symmetry excluding the trivial identity operator, and let $\hat{Y} = y^a Y^a$. The commutator is

$$i[D, Y_{\mu\sigma\nu\rho}^a(x)] = x^\alpha \partial_\alpha Y_{\mu\sigma\nu\rho}^a(x) + \gamma^{ab} Y_{\mu\sigma\nu\rho}^b(x), \quad (4.53)$$

where γ^{ab} is the matrix of anomalous dimension. This is still consistent with scale invariance as the algebra of D and P_μ is not affected due to space integration. This means that the

commutator $[D, P_\mu]$ is unchanged because the improvement terms vanish when integrated over the entire space to compute the conserved charges. In [12], Pochinski argues that in $d = 2$ we can always improve the stress tensor so that it scales according to its canonical scaling dimension as long as there is no dimension zero operator other than the identity operator. The improved stress tensor in d dimension is

$$\Theta_{\mu\nu}(x) = T_{\mu\nu}(x) + y^a(d-2-\gamma)_{ab}^{-1}\partial^\sigma\partial^\rho Y_{\mu\sigma\nu\rho}^b(x). \quad (4.54)$$

The discreteness of the scaling dimension, which we argued is true for radial quantization in chapter 2 due to compactness, and the absence of a dimension zero operator, allows us to invert the matrix so that the improved stress tensor always exists. Let us check that this improved stress tensor scales with the canonical scaling dimension

$$\begin{aligned} i[D, \Theta_{\mu\nu}] &= i[D, T_{\mu\nu}(x)] + iy^a(d-2-\gamma)_{ab}^{-1}[D, \partial^\sigma\partial^\rho Y_{\mu\sigma\nu\rho}^b(x)] \\ &= x^\alpha\partial_\alpha T_{\mu\nu}(x) + dT_{\mu\nu}(x) + y^a\partial^\sigma\partial^\rho Y_{\mu\sigma\nu\rho}^a(x) + iy^a(d-2-\gamma)_{ab}^{-1}\partial^\sigma\partial^\rho[D, Y_{\mu\sigma\nu\rho}^b(x)] \\ &= x^\alpha\partial_\alpha T_{\mu\nu}(x) + dT_{\mu\nu}(x) + y^a\partial^\sigma\partial^\rho Y_{\mu\sigma\nu\rho}^a(x) \\ &\quad + y^a(d-2-\gamma)_{ab}^{-1}\partial^\sigma\partial^\rho \left(x^\alpha\partial_\alpha Y_{\mu\sigma\nu\rho}^b(x) + \gamma^{bc}Y_{\mu\sigma\nu\rho}^c(x) \right) \\ &= x^\alpha\partial_\alpha T_{\mu\nu}(x) + dT_{\mu\nu}(x) + y^a\partial^\sigma\partial^\rho Y_{\mu\sigma\nu\rho}^a(x) \\ &\quad + y^a(d-2-\gamma)_{ab}^{-1} \left(2\partial^\sigma\partial^\rho Y_{\mu\sigma\nu\rho}^b(x) + x^\alpha\partial_\alpha\partial^\sigma\partial^\rho Y_{\mu\sigma\nu\rho}^b(x) + \gamma^{bc}\partial^\sigma\partial^\rho Y_{\mu\sigma\nu\rho}^c(x) \right) \\ &= x^\alpha\partial_\alpha T_{\mu\nu}(x) + dT_{\mu\nu}(x) + y^a\partial^\sigma\partial^\rho Y_{\mu\sigma\nu\rho}^a(x) + y^a(d-2-\gamma)_{ab}^{-1}x^\alpha\partial_\alpha\partial^\sigma\partial^\rho Y_{\mu\sigma\nu\rho}^b(x) \\ &\quad + y^a(d-2-\gamma)_{ab}^{-1}(2+\gamma)_{bc}\partial^\sigma\partial^\rho Y_{\mu\sigma\nu\rho}^c(x) \\ &= x^\alpha\partial_\alpha \left(T_{\mu\nu}(x) + y^a(d-2-\gamma)_{ab}^{-1}\partial^\sigma\partial^\rho Y_{\mu\sigma\nu\rho}^b(x) \right) + dT_{\mu\nu}(x) \\ &\quad + y^a(d-2-\gamma)_{ab}^{-1}(d-2-\gamma)_{bc}\partial^\sigma\partial^\rho Y_{\mu\sigma\nu\rho}^c(x) \\ &\quad + y^a(d-2-\gamma)_{ab}^{-1}(2+\gamma)_{bc}\partial^\sigma\partial^\rho Y_{\mu\sigma\nu\rho}^c(x) \\ &= x^\alpha\partial_\alpha \Theta_{\mu\nu}(x) + d\Theta_{\mu\nu}(x). \end{aligned}$$

This shows that the improved stress tensor scales according to the canonical scaling dimension. As all the correlation functions fall off with distance implies that the γ is positive definite. Thus, under very broad conditions of unitarity and discreteness of the spectrum of the matrix of anomalous dimension, scale invariance implies conformal invariance in two dimensions.

In this chapter, we demonstrated the conditions which scale and conformal invariance impose on the trace of the stress tensor and the conditions under which scale invariance implies conformal invariance. We showed a few examples of the theories that were scale invariant but not necessarily conformally invariant. The $U(1)$ Maxwell theory is scale invariant in all dimensions but conformally invariant only in $d = 4$. We then showed a rigorous proof of why scale invariance implies conformal invariance in $d = 2$, this time while considering the additional terms in the stress tensor due to the renormalization group. This proof relied heavily on compactness and unitarity. Thus, we demonstrated that the relation between scale and conformal invariance is not trivial. In the next chapter, we will demonstrate a counterexample in $d = 2$, which is the nonlinear sigma model.

Chapter 5

Non-Linear Sigma Model

5.1 Introduction

In this chapter, we will explore the counterexample due to Hull and Townsend [7], which is the two-dimensional nonlinear sigma model. The nonlinear sigma model is a field theory whose field takes values in the Riemannian manifold. The fields are the coordinates in the internal manifold whose metric is field-dependent. The study of the nonlinear sigma model is due to the importance of symmetry principles in physics. In this chapter, we will demonstrate through the calculation of the one-loop Weyl anomaly in the 2D nonlinear sigma model that scale invariance does not imply conformal invariance at one-loop. We will use the background field expansion method and Riemann normal coordinates to obtain a covariant expansion of the classical action. This will enable us to calculate the one-loop trace of the stress tensor following the method described in [2].

5.2 Conformally Invariant Field Theory in Two Dimensions

In this section, we will describe the properties of the conformally invariant field theories in two-dimensions [1]. In this section, we will look at the properties of the Nambu-Goto action and the Polyakov action and why it is better to work with the Polyakov action, even though both give rise to the same equations of motion. We will look at the symmetries associated with the Polyakov action and how this symmetry becomes anomalous at the quantum level.

5.2.1 The Nambu-Goto and Polyakov Action

A zero-dimensional point traces out a world line in spacetime, in the same way a worldsheet represents a one-dimensional string. The Nambu-Goto action describes a string propagating in a D -dimensional spacetime. The simplest action that we can write down is the invariant area of the worldsheet. Consider a D -dimensional spacetime. Let $\{\xi^a\}$ with $a = 1, 2$, be the parameters of this world sheet. The worldsheet is described using the functions that map position in parameter space $\{\xi^a\}$ to a point in spacetime, for each value of $\{\xi^a\}$ these functions specify a unique vector in the D -dimensional space time

$$X(\xi) \equiv (X^1(\{\xi\}), \dots, X^D(\{\xi\})). \quad (5.1)$$

These mappings describe the shape of the world sheet. The choice of coordinates on the worldsheet is not unique, so different Lorentz observers will disagree on the coordinates they assign to a particular point. But the proper area of the worldsheet should be invariant for all observers. The induced metric on the world sheet is the pull-back of the flat Minkowski space metric [14]

$$g_{ab} = \frac{\partial X^\mu}{\partial \xi^a} \frac{\partial X^\nu}{\partial \xi^b} \eta_{\mu\nu}. \quad (5.2)$$

The metric has the form

$$g_{ab} = \begin{pmatrix} \dot{X}^2 & \dot{X} \cdot X' \\ \dot{X} \cdot X' & X'^2 \end{pmatrix}, \quad (5.3)$$

where $\dot{X} = \partial X / \partial \xi^1$ and $X' = \partial X / \partial \xi^2$. The determinant is $\det(g) = \sqrt{\dot{X}^2 X'^2 - (\dot{X} \cdot X')^2}$, so the action is

$$S_{NG} = \frac{1}{2\pi\alpha'} \int d^2\xi \sqrt{\dot{X}^2 X'^2 - (\dot{X} \cdot X')^2}. \quad (5.4)$$

This is the Nambu-Goto action. The parameter α' appears in the action from dimensional analysis. α' plays the role of a coupling constant, and we will use it to keep track of loop orders in the perturbation theory. Now we will derive the equations of motion for (5.4), we will first define the conjugate momenta Π of the fields as

$$\begin{aligned} \Pi_\mu^\tau &= \frac{\partial \mathcal{L}}{\partial \dot{X}^\mu} \propto \frac{\dot{X}_\mu (X'^2) - X'_\mu (\dot{X} \cdot X')}{\sqrt{\dot{X}^2 X'^2 - (\dot{X} \cdot X')^2}}, \\ \Pi_\mu^\sigma &= \frac{\partial \mathcal{L}}{\partial X'^\mu} \propto \frac{X'_\mu (\dot{X}^2) - \dot{X}_\mu (\dot{X} \cdot X')}{\sqrt{\dot{X}^2 X'^2 - (\dot{X} \cdot X')^2}}. \end{aligned} \quad (5.5)$$

The equations of motion are given as

$$\partial_a \Pi_\mu^a = 0. \quad (5.6)$$

The equations of motion are nonlinear, and quantization becomes difficult. The Nambu-Goto action (5.4) is just

$$S_{NG} = \frac{1}{2\pi\alpha'} \int d^2\xi \sqrt{g}. \quad (5.7)$$

The variation of action under $g_{ab} \rightarrow g_{ab} + \delta g_{ab}$ and using the definition of the induced metric (5.2) together with the fact that $\delta g = \frac{1}{2} g^{ab} \delta g_{ab}$ we get the equation of motion as

$$\partial_b \left(\sqrt{g} g^{ab} \partial_a X^\mu \right) = 0. \quad (5.8)$$

If we write the above equation in terms of X^μ , we get the same nonlinear equations of motion as above.

We know from quantum field theory [11] that actions that are quadratic in derivatives are relatively easier to quantize. There is another action which is classically equivalent to the Nambu-Goto action (5.4)

$$S_P = \frac{1}{4\pi\alpha'} \int d^2\xi \sqrt{\gamma} \gamma^{ab} \partial_a X^\mu \partial_b X^\nu G_{\mu\nu}(X). \quad (5.9)$$

This is the Polyakov action. The worldsheet metric appears as an independent dynamical variable, unlike in the Nambu-Goto action (5.4) where it was the induced metric on the worldsheet. The variation of the Polyakov action (5.9) under $X^\mu \rightarrow X^\mu + \delta X^\mu$ gives the equation of motion of the fields as

$$\partial_b \left(\sqrt{\gamma} \gamma^{ab} \partial_a X^\mu \right) = 0. \quad (5.10)$$

The equations of motion of X^μ are the same as those obtained using the Nambu-Goto action, except that γ_{ab} is an independent dynamical variable which is fixed by its equations of motion. The equations of motion of the dynamical field X^μ are

$$\partial_b \left(\sqrt{\gamma} \gamma^{ab} \partial_a X^\mu \right) = 0. \quad (5.11)$$

This is the same as the equation of motion of the Nambu-Goto action. For simplicity, we will assume that the spacetime is flat $G_{\mu\nu} = \eta_{\mu\nu}$. The variation $\gamma_{ab} = \gamma_{ab} + \delta\gamma_{ab}$ fixes γ_{ab} as [14]

$$\begin{aligned}\gamma_{ab} &= 2f(\{\xi\})\partial_a X^\mu \partial_b X_\mu \\ \Rightarrow f^{-1} &= \gamma^{ab}\partial_a X^\mu \partial_b X_\mu.\end{aligned}\tag{5.12}$$

This tells us that the metric γ_{ab} is not the same as the induced metric g_{ab} . The two metrics differ by a conformal factor $f(\xi)$. This is not a problem as in two dimensions, the product of the square root of the metric and the contravariant component of the metric is independent of such factors. The actions (5.4) and (5.9) are defined on the worldsheet, which is a two-dimensional manifold, so apart from Poincaré invariance, they should also be invariant under worldsheet parametrization (diffeomorphism invariant) as the physics should not depend on the particular choice of coordinates. A diffeomorphism is a change of coordinates on the worldsheet from $\xi \rightarrow f^a(\xi)$, under such a change, the notion of distance or the value of the field at a point does not change. The immediate consequence of parametrization invariance is the conservation of the stress tensor. To see this, consider the infinitesimal reparametrization

$$\xi^a \rightarrow \xi^a + v^a(\xi).\tag{5.13}$$

Under such infinitesimal reparametrization, it follows that

$$\begin{aligned}g'^{ab} &= g^{ab} - (\nabla^a v^b + \nabla^b v^a), \\ X'^\mu &= X^\mu + v^a \nabla_a X^\mu.\end{aligned}\tag{5.14}$$

Let $A(g, X)$ be the action, then requiring that the action be invariant under this reparametrization implies

$$A(g', X') - A(g, X) = 0.\tag{5.15}$$

This gives us

$$\nabla^a T_{ab} = \frac{2\pi}{\sqrt{g}} \frac{\delta A(g, X)}{\delta X^\mu} \nabla_b X^\mu,\tag{5.16}$$

where

$$T_{ab} = \frac{4\pi}{\sqrt{g}} \frac{\delta A(g, X)}{\delta g^{ab}}\tag{5.17}$$

The right-hand side of the above equation is zero on-shell, so we get $\nabla^a T_{ab} = 0$. For the Polyakov action

$$T_{ab} = \frac{1}{\alpha'} G_{\mu\nu}(X) \left(\partial_a X^\mu \partial_b X^\nu - \frac{1}{2} g_{ab} \partial^c X^\mu \partial_c X^\nu \right).\tag{5.18}$$

The term in the round bracket on the right-hand side of the above equation is exactly what we set to zero to get the equation of motion of the dynamical field γ_{ab} in the Polyakov action (5.9). Thus setting $T_{ab} = 0$ gets rid of extra degrees of freedom associated with γ_{ab} . The physics of the strings should be completely described by the fields X^μ . Thus, we have a two-dimensional surface, and on it we have scalar fields X^μ . We will show that the physics is independent of the two-dimensional metric γ_{ab} .

5.2.2 Weyl and Conformal Invariance

In the previous section, we showed that to eliminate the extra degrees of freedom of the Polyakov action (5.9), we require that all the components of the stress tensor must vanish. This means we expect three independent equations, as the stress tensor is symmetric. We can see from (5.18) that at the classical level the trace vanishes. But the trace does not necessarily vanish when we quantize the theory. Thus, the classical vanishing of the trace is anomalous,

and we need to ensure that the trace vanishes at the quantum level as well, which corresponds to, as we demonstrated in chapter 4, conformal invariance in the two-dimensional theory. We have already discussed the reparametrization invariance of two-dimensional theories in the previous section. In this section, we will talk about Weyl transformation and conformal reparametrization. At the end of this section, we will show how these two are related to conformal transformation.

In the previous section, we already argued that the Polyakov action (5.9) is not affected by conformal factors present with the metric. This means that the Polyakov action is invariant under local rescaling of the metric, which preserves angles. This is called the Weyl transformation, and it acts in the following way

$$\begin{aligned}\delta\gamma_{ab} &= -\delta\phi(\xi)\gamma_{ab}, \\ \delta X^\mu &= 0.\end{aligned}\tag{5.19}$$

This is the infinitesimal version of the Weyl transformation. The finite version is

$$\gamma_{ab} \rightarrow \exp(\phi(\xi))\gamma_{ab}.\tag{5.20}$$

If a general action $A(g, X)$ is invariant under Weyl transformation, it follows that the trace of the stress tensor vanishes

$$\frac{\delta A(g, X)}{\delta\phi} = g^{ab}T_{ab} \Rightarrow T^a_a = 0.\tag{5.21}$$

Two metrics g_1 and g_2 on a real two-dimensional manifold are said to be conformally equivalent if there exists a scalar function $\phi(\xi)$ such that $g_2 = \exp(\phi(\xi))g_1$. This equivalence relation partitions the space of metrics on the manifold into equivalence classes called conformal classes [8]. Thus, the sole effect of the Weyl transformation is to move the metric around in its conformal classes.

Reparametrization invariance (5.13) allows us to choose the two functions $v^a(\xi)$ in a way so that the two-dimensional metric has only one degree of freedom rather than three. A convenient choice is to choose coordinates in such a way that the metric becomes conformally Euclidean

$$\gamma_{ab} = \exp(\phi(\xi))\delta_{ab}.\tag{5.22}$$

This represents a Weyl transformation on a flat metric. As we have seen in previous chapters, it is better to work in complex coordinates, so the line element becomes

$$ds^2 = \exp(\phi)dzd\bar{z},\tag{5.23}$$

so that the metric is just

$$\gamma_{ab} = \begin{pmatrix} 0 & \frac{e^\phi}{2} \\ \frac{e^\phi}{2} & 0 \end{pmatrix}.\tag{5.24}$$

An interesting property of the conformal gauge is that there are no mixed components in Christoffel connections. The only non-vanishing components are

$$\Gamma_{zz}^z = \partial_z\phi \quad \text{and} \quad \Gamma_{\bar{z}\bar{z}}^{\bar{z}} = \partial_{\bar{z}}\phi.\tag{5.25}$$

The Ricci scalar R in this coordinate system is given by

$$R = -4e^\phi\partial_z\partial_{\bar{z}}\phi.\tag{5.26}$$

Now, consider a general coordinate transformation

$$z \rightarrow z + v^z(z, \bar{z}) \quad \text{and} \quad \bar{z} \rightarrow \bar{z} + v^{\bar{z}}(z, \bar{z}),\tag{5.27}$$

such a coordinate, even if we start with a conformal gauge, will take us out of it, as can be seen as follows

$$\begin{aligned}\delta\gamma^{zz} &= -2\nabla^z v^z, \\ \delta\gamma^{\bar{z}\bar{z}} &= -2\nabla^{\bar{z}} v^{\bar{z}}, \\ \delta\gamma^{z\bar{z}} &= -\nabla^z v^{\bar{z}} - \nabla^{\bar{z}} v^z.\end{aligned}\tag{5.28}$$

So we consider a special subset of such transformations called conformal reparametrization, such that

$$\partial_z v^z = \partial_{\bar{z}} v^{\bar{z}} = 0.\tag{5.29}$$

A conformal reparametrization is a coordinate transformation for which v^z is a function of z alone (analytic) and $v^{\bar{z}}$ is a function of $\bar{z}$ alone (anti-analytic). The transformation generated by such reparametrizations is

$$\begin{aligned}\delta\gamma^{zz} &= \delta\gamma^{\bar{z}\bar{z}} = 0, \\ \delta\gamma^{z\bar{z}} &= -\gamma^{z\bar{z}}\delta\phi(z, \bar{z}), \quad \delta\phi(z, \bar{z}) = \nabla_z v^z + \nabla_{\bar{z}} v^{\bar{z}}, \\ \delta X^\mu &= v^z \nabla_z X^\mu + v^{\bar{z}} \nabla_{\bar{z}} X^\mu.\end{aligned}\tag{5.30}$$

A conformal transformation is a dynamical transformation of only the fields, it leaves the metric invariant.

$$\begin{aligned}\delta\gamma^{zz} &= \delta\gamma^{\bar{z}\bar{z}} = \delta\gamma^{z\bar{z}} = 0, \\ \delta X^\mu &= v^z \nabla_z X^\mu + v^{\bar{z}} \nabla_{\bar{z}} X^\mu.\end{aligned}\tag{5.31}$$

From the above definitions, it is clear that if we have a conformal transformation which acts only on the fields X^μ together with an appropriately chosen Weyl transformation of the metric, we have conformal reparametrization. In this section, we showed how, classically under the described transformations, the trace of the stress tensor vanishes, which is in agreement with what we showed in 4, that for conformal invariance the trace should vanish. In the next section, we will show that this classical vanishing of the trace is anomalous at the quantum level.

5.2.3 The Quantum Anomaly

The discussion of reparametrization, Weyl transformation, and conformal invariance in the previous sections was in the context of classical field theory. In this section, we will discuss these transformations for the path integral formulation [8]. The partition function for a general action $A(g, X)$ in two dimensions is

$$Z[\gamma] = \exp -W[\gamma] \equiv \int [DX] \exp (-A(g, X)),\tag{5.32}$$

where $[DX]$ is the functional integral over all configurations of the fields X . The partition function is reparametrization invariant if

$$Z[\gamma'] = Z[\gamma],\tag{5.33}$$

for all γ' and γ related by reparametrization (5.14). This is only true when the functional measure and the action are reparametrization invariant. For the action to be parametrization invariant, we require that the classical action and the counterterms are both reparametrization invariant. This means that we have chosen a renormalization scheme that is reparametrization invariant, like dimensional regularization.

The classical action depends explicitly on γ_{ab} , but the functional integration measure $\int [DX] \exp(-A(g, X))$ implicitly depends on it. The functional integration measure involves computing $\exp(-A(g, X))$ for each X and then summing over all X . The measure DX is the volume of each infinitesimal volume element. It counts the number of nearby configurations for which the value of $\exp(-A(g, X))$ is the same. So we need a counting procedure that is reparametrization invariant because we have already assumed that the effective quantum action is reparametrization invariant. We need a notion of distance on the configuration space that is reparametrization invariant. The distance between two nearby configurations X and $X + \delta X$ is given by

$$||\delta X||_{\gamma, X}^2 = \int d^2\zeta \sqrt{\gamma} \delta X^\mu \delta X^\nu G_{\mu\nu}(X). \quad (5.34)$$

We have already said that reparametrization does not change the value of the scalar field and the notion of distance, so the (5.34) is reparametrization invariant. That is if X' and $X' + \delta X'$ are obtained by reparametrization to X and $X + \delta X$ then

$$||\delta X'||_{\gamma', X'} = ||\delta X||_{\gamma, X}. \quad (5.35)$$

This ensures that the counting procedure for the number of configurations is reparametrization invariant. Now that we have defined in a reparametrization invariant way the quantum action and the procedure to calculate the volume in configuration space, the partition function must also be reparametrization invariant. This implies

$$\int d^2\zeta \frac{\delta W}{\delta \gamma^{ab}} \delta \gamma^{ab} = 0, \quad (5.36)$$

where $\delta \gamma^{ab} = -\nabla^a v^b - \nabla^b v^a$. This equation can be rewritten in conformal gauge in $z\bar{z}$ coordinates as

$$\begin{aligned} 0 &= \int d^2z \left[2 \frac{\delta W}{\delta \gamma^{zz}} \nabla^z v^z + 2 \frac{\delta W}{\delta \gamma^{\bar{z}\bar{z}}} \nabla^{\bar{z}} v^{\bar{z}} - \frac{\delta W}{\delta \phi} (\nabla_z v^z + \nabla_{\bar{z}} v^{\bar{z}}) \right] \\ 0 &= \int d^2z \sqrt{\gamma} \left\{ v^z \left(2 \nabla^z \frac{1}{\sqrt{\gamma}} \frac{\delta}{\delta \gamma^{zz}} - \nabla_z \frac{1}{\sqrt{\gamma}} \frac{\delta}{\delta \phi} \right) W[\gamma] + v^{\bar{z}} \left(2 \nabla^{\bar{z}} \frac{1}{\sqrt{\gamma}} \frac{\delta}{\delta \gamma^{\bar{z}\bar{z}}} - \nabla_{\bar{z}} \frac{1}{\sqrt{\gamma}} \frac{\delta}{\delta \phi} \right) W[\gamma] \right\} \end{aligned} \quad (5.37)$$

Now since $v^z(v^{\bar{z}})$ is arbitrary we get

$$\nabla_z \left(2 \nabla^z \frac{1}{\sqrt{\gamma}} \frac{\delta W}{\delta \gamma^{zz}} \right) = \nabla_z \left(\frac{1}{\sqrt{\gamma}} \frac{\delta W}{\delta \phi} \right) \quad (5.38)$$

and a similar equation corresponding to $\bar{z}$. Now using (5.17) we can identify the left-hand side of the above equation as $\nabla^z \langle T_{zz} \rangle$. Classically, we then expect the right-hand side of the above equation to be zero for a conformally invariant theory. However, this is not true quantum mechanically. If we assume that the right-hand side is a local function on the worldsheet, we can determine its structure. The left-hand side is a tensor of type t_z under conformal reparametrization and is of scaling dimension one on the worldsheet. The only function of γ_{ab} that has these properties is the z -derivative of the scalar curvature R . Thus, the assumption of locality, together with dimensional analysis, tells us that

$$\nabla_z \left(\frac{1}{\sqrt{\gamma}} \frac{\delta W}{\delta \phi} \right) = \frac{\lambda}{48\pi} \nabla_z R. \quad (5.39)$$

The proportionality constant λ cannot be determined by general arguments and is specific to the theory. The stress tensor for a conformally invariant field theory can be written in terms

of the moments $L_n(\bar{L}_n)$, which satisfy the Virasoro algebra. The constant λ is nothing but the central charge of the theory. Using the above equation, we get

$$\frac{\delta W}{\delta \phi} = \frac{\lambda}{48\pi} \sqrt{\gamma} (R + \mu^2), \quad (5.40)$$

where μ^2 is a constant. We know that the conformal scale factor always refers the metric to some given reference metric

$$\gamma_{ab} = e^\phi \hat{\gamma}_{ab}. \quad (5.41)$$

In conformal gauge $\hat{\gamma}_{ab} = \delta_{ab}$. Then using (5.26) we get

$$W = \frac{\lambda}{48\pi} \int d^2\zeta \left(\frac{1}{2} \hat{\gamma}^{ab} \partial_a \phi \partial_b \phi + \mu^2 e^\phi \right) + \text{conformally invariant terms.} \quad (5.42)$$

Thus, the assumption that the anomaly is local has helped us identify the conformally non-invariant part of the quantum effective action. The first term on the right-hand side of the above equation is the Liouville action. This term depends on the choice of the metric on the two-dimensional worldsheet.

5.3 Background Field Expansion and Riemann Normal Coordinates

In the previous section, we saw the symmetries associated with the Polyakov action. In particular, we demonstrated the meaning of the reparametrization invariance at both the classical and quantum levels. In this section, we set up the path integral perturbation theory for the Polyakov action (5.9). Thus, it will be useful if we set up the perturbation expansion in such a way that these symmetries are manifest. This means we want to calculate the Feynman diagrams using covariant vertices and propagators. The regularization scheme for divergent integrals should be compatible with the general covariance of the spacetime. This way, any counterterms will be covariant, and any anomalies encountered will be expressed in a coordinate-invariant way. This can be achieved by using the background field expansion method. In this method, we separate the two-dimensional field into two parts, a background or classical part, and a quantum part

$$X^\mu(\zeta) = X_0^\mu(\zeta) + \pi^\mu(\zeta), \quad (5.43)$$

and then shift the path integral measure $[DX]_\gamma \rightarrow [D\pi]_\gamma$, this is why π are called quantum fields. The background partition function is defined as [2]

$$\Omega[X_0, \gamma] = \int [D\pi] e^{-\{A[X_0 + \pi] - A[X_0] - \int d^2\zeta \frac{\delta A}{\delta X_0^\mu(\zeta)} \pi^\mu(\zeta)\}}. \quad (5.44)$$

In the above partition function, we subtract the $A[X_0]$ part of the action because it is not integrated over and is just a constant. At the same time, $\delta A / \delta X_0^\mu(\zeta)$ is just the equation of motion of the classical field, which vanishes classically. Next we expand the action $A[X_0 + \pi]$ in powers of the quantum field π^μ and derive the Feynman rules. From quantum field theory [11] we know that the propagator is given by the quadratic term, in this case, the propagator will also depend on the classical fields X_0 . Although this expansion is valid, it is not manifestly spacetime covariant. This is because the quantum field π^μ is the coordinate difference between the full field X^μ and the classical background field X_0^μ and therefore will not transform as a vector under coordinate transformations. So we need to replace π^μ with a spacetime vector, a natural choice is the tangent vector to the geodesic connecting X_0^μ and $X_0^\mu + \pi^\mu$. Let us assume that there is unique geodesic $\lambda^\mu(t)$ with affine parameter t such that $\lambda^\mu(0) = X_0^\mu(\zeta)$ and $\lambda^\mu(1) = X_0^\mu(\zeta) + \pi^\mu(\zeta)$. Let η^μ be the tangent to $\lambda^\mu(t)$ at X_0^μ , i.e.,

$\eta^\mu = \dot{\lambda}^\mu(0)$. The geodesic equation is

$$\frac{d^2\lambda^\mu(t)}{dt^2} + \Gamma_{\alpha\beta}^\mu \frac{d\lambda^\alpha(t)}{dt} \frac{d\lambda^\beta(t)}{dt} = 0. \quad (5.45)$$

We expand $\lambda^\mu(t)$ about $t = 0$, the first few terms are

$$\begin{aligned} \lambda^\mu(t) &= \lambda^\mu(0) + \dot{\lambda}^\mu(0)t + \frac{1}{2}\ddot{\lambda}^\mu(t)t^2 + \dots \\ &= X_0^\mu(\xi) + \eta^\mu t - \frac{1}{2}(\Gamma_{\alpha\beta}^\mu)_0 \dot{\lambda}^\alpha(0)\dot{\lambda}^\beta(0)t^2 + \dots \end{aligned} \quad (5.46)$$

The higher-order terms in the expansion can be calculated using (5.45) by repeated differentiation. If we consider the next leading order term in the above equation, differentiate (5.45) with respect to parameter t . This gives

$$\begin{aligned} \ddot{\lambda}^\mu(t) &= -\Gamma_{\alpha\beta,\gamma}^\mu \dot{\lambda}^\alpha(t)\dot{\lambda}^\beta(t)\dot{\lambda}^\gamma(t) - \Gamma_{\alpha\beta}^\mu \ddot{\lambda}^\alpha(t)\dot{\lambda}^\beta(t) - \Gamma_{\alpha\beta}^\mu \dot{\lambda}^\alpha(t)\ddot{\lambda}^\beta(t) \\ &= -\nabla_\gamma \Gamma_{\alpha\beta}^\mu \dot{\lambda}^\alpha(t)\dot{\lambda}^\beta(t)\dot{\lambda}^\gamma(t). \end{aligned} \quad (5.47)$$

The last step in the above derivation follows from using (5.45) for $\ddot{\lambda}^\alpha(t)$. Thus, the expansion becomes

$$\lambda^\mu(t) = X_0^\mu(\xi) + \eta^\mu t - \frac{1}{2}\Gamma_{\alpha\beta}^\mu \eta^\alpha \eta^\beta t^2 - \frac{1}{3!}\Gamma_{\alpha\beta\gamma}^\mu \eta^\alpha \eta^\beta \eta^\gamma t^3 + \dots \quad (5.48)$$

Let $y^\mu = \eta^\mu t$ then,

$$\lambda^\mu(t) = X_0^\mu(\xi) + y^\mu - \frac{1}{2}\Gamma_{\alpha\beta}^\mu y^\alpha y^\beta - \frac{1}{3!}\Gamma_{\alpha\beta\gamma}^\mu y^\alpha y^\beta y^\gamma + \dots, \quad (5.49)$$

here $\Gamma_{\alpha\beta\gamma}^\mu$ stands for the formal covariant derivative only on the lower indices. As the above equation does not involve η^μ it is valid for all geodesic passing through X_0^μ . This defines an invertible coordinate transformation between $X_0^\mu + \pi^\mu$ in the initial coordinate chart near X_0^μ and the new coordinates y^μ as

$$\left| \frac{\partial \lambda}{\partial y} \right| \neq 0, \quad (5.50)$$

for a given set of η^μ , $y^\mu = \eta^\mu t$, define the equation of a geodesic in the new coordinate, called Riemann normal coordinates. In this coordinate, the equation of a geodesic is a straight line. This method is valid in the neighborhood of X_0^μ when two geodesics do not meet again in the domain and the coordinate transformation is invertible [4]. At $t = 1$ we have $\lambda^\mu(1) = X_0^\mu(\xi) + \pi^\mu(\xi)$, using this, the quantum field can be expressed in the Riemann normal coordinates as

$$\pi^\mu = y^\mu - \frac{1}{2}\Gamma_{\alpha\beta}^\mu y^\alpha y^\beta + \dots \quad (5.51)$$

The benefit of using the Riemann normal coordinate is the fact that in these coordinates the equation of a geodesic is a straight line $y^\mu = \eta^\mu t$. As this must also satisfy (5.45), this tells us that in these coordinates all components of the Christoffel symbol vanish. Also the all the higher-order Christoffel symbols vanish when symmetrized in the lower indices,

$$\begin{aligned} \Gamma_{bc}^a &= 0, \\ \Gamma_{(bc\dots)}^a &= 0. \end{aligned} \quad (5.52)$$

This simplifies the expression for the curvature tensor, the equation for the Riemann tensor is given as

$$\begin{aligned} R_{bcd}^a &= \partial_c \Gamma_{bd}^a - \partial_d \Gamma_{bc}^a + \Gamma_{kc}^a \Gamma_{bd}^k - \Gamma_{kd}^a \Gamma_{bc}^k \\ &= \partial_c \Gamma_{bd}^a - \partial_d \Gamma_{bc}^a. \end{aligned} \quad (5.53)$$

The last two terms vanish in the Riemann normal coordinates. Consider $\Gamma_{(bcd)}^a = 0$

$$\begin{aligned} \Gamma_{bc,d}^a + \Gamma_{cd,b}^a + \Gamma_{db,c}^a &= 0 \\ \Gamma_{bc,d}^a &= \frac{1}{3} (R_{bdc}^a + R_{cdb}^a). \end{aligned} \quad (5.54)$$

The Taylor expansion in normal coordinates is

$$T_{\mu_1 \mu_2 \dots} = \sum_i \frac{1}{i!} (\partial_{\nu_1} \dots \partial_{\nu_i} T_{\mu_1 \mu_2 \dots}) \eta^{\nu_1} \dots \eta^{\nu_i}. \quad (5.55)$$

Using the relations derived above, we can write the derivatives in Riemann normal coordinates in a covariant way. Up to second order, the expansion of a rank-2 covariant tensor can be written as

$$\begin{aligned} T_{\mu\nu}(X_0 + \pi) &= T_{\mu\nu}(X_0) + \nabla_\lambda T_{\mu\nu}(X_0) \eta^\lambda \\ &+ \frac{1}{2} \left(\nabla_\lambda \nabla_\sigma T_{\mu\nu}(X_0) - \frac{1}{3} R_{\lambda\mu\sigma}^\rho T_{\rho\nu}(X_0) - \frac{1}{3} R_{\lambda\nu\sigma}^\rho T_{\mu\rho}(X_0) \right) \eta^\lambda \eta^\sigma + \dots \end{aligned} \quad (5.56)$$

The above expression contains only covariant derivatives, so the expansion is manifestly covariant under general coordinate transformations. Let us now consider the expansion of terms involved in the nonlinear sigma model. Using the above expansion, we can write the expansion of the spacetime metric

$$G_{\mu\nu}(X_0 + \pi) = G_{\mu\nu}(X_0) + \frac{1}{3} R_{\mu\lambda\nu\sigma} \eta^\lambda \eta^\sigma + \dots \quad (5.57)$$

In the derivation of the expansion of the spacetime metric, we used the fact that the covariant derivative of the metric is zero and the symmetry properties of the Riemann tensor. Now using (5.51) we can find the expansion of $\partial_a(X_0^\mu + \pi^\mu)$ in the (5.9).

$$\partial_a(X_0^\mu + \pi^\mu) = \partial_a X_0^\mu + \nabla_a \eta^\mu + \frac{1}{3} R_{\lambda\sigma\nu}^\mu \partial_a X_0^\nu \eta^\lambda \eta^\sigma + \dots, \quad (5.58)$$

where $\nabla_a \eta^\mu = \partial_a \eta^\mu + \Gamma_{\lambda\sigma}^\mu \partial_a X_0^\lambda \eta^\sigma$. Using all of the above results up to quadratic order in η , the expansion of the Polyakov action (5.9) is

$$\begin{aligned} S_P[X_0 + \pi] &= S[X_0] + \frac{1}{2\pi\alpha'} \int d^2\xi \sqrt{\gamma} \gamma^{ab} G_{\mu\nu}(X_0) \partial_a X_0^\mu \nabla_b \eta^\nu \\ &+ \frac{1}{4\pi\alpha'} \int d^2\xi \sqrt{\gamma} \gamma^{ab} \left\{ G_{\mu\nu}(X_0) \nabla_a \eta^\mu \nabla_b \eta^\nu + R_{\mu\lambda\sigma\nu}(X_0) \partial_a X_0^\mu \partial_b X_0^\nu \eta^\lambda \eta^\sigma \right\} \\ &+ \frac{1}{3\pi\alpha'} \int d^2\xi \sqrt{\gamma} \gamma^{ab} R_{\mu\lambda\sigma\nu}(X_0) \partial_a X_0^\mu \eta^\lambda \eta^\sigma \eta^\nu \\ &+ \frac{1}{12\pi\alpha'} \int d^2\xi \sqrt{\gamma} \gamma^{ab} R_{\mu\lambda\sigma\nu} \eta^\lambda \eta^\sigma \partial_a \eta^\nu \partial_b \eta^\nu + \dots \end{aligned} \quad (5.59)$$

We are free to choose the configuration of the background field, we can choose it in such a way that it satisfies the equations of motion of the Polyakov action (5.9) and so the linear

term in η vanishes. The second term (5.59) contains a term quadratic in derivatives, which is the kinetic term. The kinetic term contains the spacetime metric, which is a function of the background fields, so the propagator would be nontrivial. To derive it, first we introduce vielbein formalism [3], which is necessary to describe spinor fields in curved manifolds. Under a change of coordinates $x \rightarrow x'$, the coordinate differential transforms as

$$dx'^{\mu} = \frac{\partial x'^{\mu}}{\partial x^{\nu}} dx^{\nu}. \quad (5.60)$$

The only requirement is that the Jacobian matrix be invertible, it should be an element of the group $GL(d)$. But local fields are defined according to their transformation properties in Euclidean or Minkowski space, where the symmetry group is $SO(d)$ or $SO(d-1,1)$ respectively. Thus, to carry the Lorentz group formalism over to general manifolds, we introduce at each point a local orthogonal frame of basis vectors

$$e^a = e_{\mu}^a dx^{\mu}, \quad (5.61)$$

where the frame vectors e^a form a tetrad. The upper index is the Lorentz index, and the lower index is the Einstein index. The orthogonality conditions determine a natural choice of these basis vectors

$$e_{\mu}^a e_{\nu}^b g^{\mu\nu} = \eta^{ab}. \quad (5.62)$$

To compare the fields at two different nearby positions, we need a notion of parallel transport which is specified by the spin connection ω_{μ}^{ab}

$$V^a \rightarrow V^a - \omega_{\mu}^{ab} dx^{\mu} V^b, \quad (5.63)$$

where dx^{μ} is the amount of parallel transport. This allows us to write the covariant derivative

$$(D_{\mu} V)^a = \partial_{\mu} V^a + \omega_{\mu}^{ab} V^b. \quad (5.64)$$

Parallel transport only changes the direction of the vector and not its magnitude, the spin connection is antisymmetric in Lorentz indices $\omega_{\mu}^{ab} = -\omega_{\mu}^{ba}$. Now referring to the local Lorentz frame, the fields η^{μ} in (5.59)

$$\eta^i = e_{\mu}^i \eta^{\mu}. \quad (5.65)$$

In this coordinate system, the kinetic term is diagonal because of the orthogonality condition

$$G_{\mu\nu}(X_0) \nabla_a \eta^{\mu} \nabla_b \eta^{\nu} = (\nabla_a \eta)^i (\nabla_b \eta)^j. \quad (5.66)$$

The form of the propagator will be straightforward in this coordinate system, so the change of variables in the path integral and integrating over the η^i fields in the local Lorentz frame is justified. As the path integral measure is defined in a spacetime-invariant manner, the change of variables does not affect it.

5.4 One Loop Calculation of Weyl Anomaly

In the previous section, we set up everything required for the computation of the one-loop anomaly. In this section, we will compute the one-loop Weyl anomaly of the sigma model using the covariant background field method and the conservation of the stress tensor. We are calculating the variation of the effective action with respect to the scale factor of the worldsheet metric. The conservation equation for the stress tensor takes the form

$$\nabla^z \langle T_{zz} \rangle + \nabla^{\bar{z}} \langle T_{\bar{z}\bar{z}} \rangle = 0 \quad (5.67)$$

In 3, we showed that in two-dimensional renormalizable quantum field theories, the components of the stress tensor are protected by symmetry and therefore renormalizations of the stress tensor components are absent. This means we can use the above equation to compute $\langle T_{z\bar{z}} \rangle$, so the value of $\langle T_{zz} \rangle$ is such that the value of $\langle T_{z\bar{z}} \rangle$ cannot be zero. To calculate $\langle T_{zz} \rangle$, we will assume that the worldsheet metric is flat and has Minkowski signature. In momentum space, the conservation equation reads

$$q_- \langle T_{++} \rangle + q_+ \langle T_{--} \rangle = 0. \quad (5.68)$$

q_- and q_+ are the light cone components defined as $q_+ = q_1 + q_0$ and $q_- = q_0 - q_1$. We expand the T_{zz} component of the stress tensor (5.18) in terms of the field η^i . We showed in the previous section that $X^\mu = X_0^\mu + e_i^\mu \eta^i + \dots$ and

$$T_{zz} = \frac{1}{\alpha'} G_{\mu\nu}(X_0) \nabla_z X^\mu \nabla_z X^\nu. \quad (5.69)$$

Substituting for X^μ in the above equation and considering terms only quadratic in $\nabla_z \eta^i$ we get

$$T_{zz} = \frac{1}{\alpha'} G_{\mu\nu}(X_0) \nabla_z X_0^\mu \nabla_z X_0^\nu + \frac{2}{\alpha'} \delta_{ij} \nabla_z X_0^\mu e_\mu^i \nabla_z \eta^j + \frac{1}{\alpha'} \delta_{ij} \nabla_z \eta^i \nabla_z \eta^j + \dots, \quad (5.70)$$

where we have neglected higher-order terms, as they involve derivatives of Christoffel symbols, which can be expressed in terms of the Riemann tensor, resulting in higher-order diagrams. The term $\nabla_z \eta^i \approx \partial_z \eta^i$, as it will involve spin connections and will result in a higher-order diagram. The contribution to $\langle T_{++} \rangle$ that comes from S_p at leading order is from the following diagram

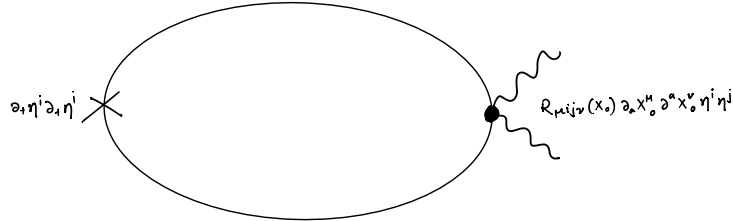


FIGURE 5.1: One-loop diagram contributing to trace. Here, the wavy lines represent the classical fields, the normal lines are the propagators, and the "x" mark represents the T_{++} insertion.

This is the only term that contributes as the first term in (5.70) contains only the classical field X_0^μ and we impose the constraint at the classical level that each term in the stress tensor must vanish to get rid of extra degrees of freedom associated with the worldsheet metric. The second term in (5.70) does not contribute at this order as there are no consistent non-vanishing Feynman diagrams. The curvature term in the (5.59) is

$$S[X_0 + \pi] \supset \int d^2 \xi R_{\mu i j \nu} \partial_a X_0^\mu \partial_b X_0^\nu \eta^i \eta^j. \quad (5.71)$$

The contribution of the above diagram is

$$\int \frac{d^2 l}{2\pi} \frac{l_+(l_+ + q_+)}{l^2(l+q)^2} [R_{\mu\nu} \partial_a X_0^\mu \partial^a X_0^\nu](q), \quad (5.72)$$

where l is the loop momentum and q is the momentum due to insertion of T_{++} . The momentum l_+ and $l_+ + q_+$ are due to Fourier transform of $\partial_+ \eta^i$, as it is a vertex factor. There is no dependence on α' as the stress tensor and action both contain $1/\alpha'$ and the propagators contain α' so it cancels out.

Now we will evaluate the integral explicitly using dimensional regularization. We first evaluate the integral in d -dimensions and then take the limit $d \rightarrow 2$. To do so, we first define a null vector $n^\mu \equiv (1, 1, 0, 0, \dots)$ in d -dimension using this we can define $l_+ = n \cdot l$ and likewise $q_+ = n \cdot q$. The integral then takes the form

$$\begin{aligned} I &= \int \frac{d^d l}{(2\pi)^{d/2}} \frac{(n \cdot l)[n \cdot (l + q)]}{l^2(l + q)^2} \\ &= \int \frac{d^d l}{(2\pi)^{d/2}} \frac{(n \cdot l)^2 + (n \cdot l)(n \cdot q)}{l^2(l + q)^2}. \end{aligned} \quad (5.73)$$

Now using Feynman parametrization [11], we can rewrite the denominator as

$$\begin{aligned} \frac{1}{l^2(l + q)^2} &= \int_0^1 dx \frac{1}{[xl^2 + (1 - x)(l + q)^2]^2} \\ &= \int_0^1 dx \frac{1}{(l^2 + 2xq \cdot l + xq^2)^2} \\ &= \int_0^1 dx \frac{1}{[(l + xq)^2 + x(1 - x)q^2]^2} \end{aligned} \quad (5.74)$$

Now the integral becomes

$$I = \int_0^1 dx \int \frac{d^d l}{(2\pi)^d} \frac{(n \cdot l)^2 + (n \cdot l)(n \cdot q)}{[(l + xq)^2 + x(1 - x)q^2]^2}. \quad (5.75)$$

We shift the integration variable $l \rightarrow l - xq$ to get

$$I = \int_0^1 dx \int \frac{d^d l}{(2\pi)^{d/2}} \frac{(n \cdot (l - qx))^2 + (n \cdot (l - qx))(n \cdot q)}{(l^2 + \Delta)^2}, \quad (5.76)$$

where $\Delta = x(1 - x)q^2$. Now we notice that the term $(n \cdot (l - qx))(n \cdot q)$ will contain a term like l^μ , this term will vanish by symmetry considerations. Now, consider the term $(n \cdot (l - qx))^2$

$$(n \cdot (l - qx))^2 = (n \cdot l)^2 - 2x(n \cdot l)(n \cdot q) + x^2(n \cdot q)^2, \quad (5.77)$$

the first term on the right-hand side of the above equation is of the form $n^\mu n^\nu l_\mu l_\nu$. The integral vanishes by symmetry unless $\mu = \nu$, Lorentz invariance therefore requires that $l_\mu l_\nu \propto l^2 g_{\mu\nu}$. Upon contraction $n^\mu n^\nu g_{\mu\nu} = 0$ as n is the null vector. So the only term that remains is

$$I = (n \cdot q)^2 \int_0^1 dx x^2 \int \frac{d^d l}{(2\pi)^{d/2}} \frac{1}{(l^2 + \Delta)^2}. \quad (5.78)$$

The integral over l is a standard integral encountered in the dimensional regularization scheme and can be evaluated easily by making a change of variables to $y = \Delta/(l^2 + \Delta)$ and then using the definition of the beta function

$$\int_0^1 dx x^{\alpha-1} (1-x)^{\beta-1} = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}. \quad (5.79)$$

The integral we are left with is

$$\begin{aligned} I &= \frac{(n \cdot q)^2 q^{d-4}}{2^d} \frac{\Gamma(2-d/2)}{\Gamma(2)} \int_0^1 dx x^2 [x(1-x)]^{d/2-2} \\ &= \frac{(n \cdot q)^2 q^{d-4}}{2^d} \frac{\Gamma(2-d/2)\Gamma(d/2-1)\Gamma(d/2+1)}{\Gamma(2)\Gamma(d)}. \end{aligned} \quad (5.80)$$

Now we substitute $d = 2 - 2\epsilon$ and then take the limit $\epsilon \rightarrow 0$ this gives us

$$I = -\frac{(n \cdot q)^2}{2q^2} (1 + \mathcal{O}(\epsilon)) \quad (5.81)$$

Now we write $n \cdot q = q_+$ and in light cone coordinates $q^2 = q_+ q_-$ using this we get

$$\int \frac{d^2 l}{2\pi} \frac{l_+ (l_+ + q_+)}{l^2 (l + q)^2} = -\frac{q_+}{2q_-}. \quad (5.82)$$

This completes the computation of $\langle T_{++} \rangle$ and using (5.68) we get

$$\langle T_{-+}(\xi) \rangle_{1-loop} = \frac{1}{2} R_{\mu\nu}(X_0) \partial_a X_0^\mu \partial^a X_0^\nu. \quad (5.83)$$

This is the conformal anomaly. The trace of the stress tensor is not zero even when we started with a conformally invariant classical action. The trace at one-loop level depends on the curvature of the spacetime. In 4, we demonstrated that the condition for scale invariance only requires that the trace of the stress tensor be a divergence of the virial current (4.14). The candidate for K^a in this case is

$$K_a = -v_\mu(X) \partial_a X^\mu, \quad (5.84)$$

where $v_\mu(X)$ is an arbitrary vector field in the spacetime and D_μ is the spacetime derivative. Then divergence $\partial^a K_a$ gives

$$\begin{aligned} \partial^a K_a &= -D_\nu v_\mu \partial_a X^\mu \partial^a X^\nu \\ &= -\frac{1}{2} (D_\mu v_\nu + D_\nu v_\mu) \partial_a X^\mu \partial^a X^\nu. \end{aligned} \quad (5.85)$$

Thus, the scale invariance condition at one-loop becomes

$$R_{\mu\nu} = D_\mu v_\nu + D_\nu v_\mu, \quad (5.86)$$

which defines a quasi-Ricci flat manifold introduced first by Friedan [5]. The condition for conformal current conservation (4.29) tells us that the condition for conformal invariance implies

$$R_{\mu\nu} = D_\mu D_\nu \Phi, \quad (5.87)$$

for some scalar field Φ in the spacetime, this can be removed by an improvement term. The scale invariance condition (5.86) allows general vector fields v_μ but the stronger conformal invariance condition only allows vector fields v_μ which are gradients of a scalar $v_\mu = D_\mu \Phi$. We can write down the improved stress tensor at one loop as

$$\Theta_{ab} = T_{ab} + \frac{1}{2} (\partial_a \partial_b - \gamma_{ab} \partial^2) \Phi(X(\xi)). \quad (5.88)$$

This implies that any compact Riemannian solution of (5.86) must be of special form (5.87). Consider the double divergence of (5.86)

$$D^\nu D^\mu R_{\mu\nu} = D^\nu D^2 v_\nu + D^\nu D^\mu D_\nu v_\mu, \quad (5.89)$$

using the fact $D^\mu R_{\mu\nu} = (1/2)D_\nu R$ and the fact that $[D_\mu, D_\nu]v^\mu = R_{\nu\rho}v^\rho$ we get

$$D^2 R + v \cdot DR = -R_{\mu\nu}R^{\mu\nu}. \quad (5.90)$$

Let us denote by $\mathcal{M}$ the target space of the sigma model (spacetime), the trace of (5.86) gives us $R = 2D \cdot v$. Integrating the trace over the sigma model target manifold, we get

$$\int_{\mathcal{M}} \sqrt{G} R d^d X = 0. \quad (5.91)$$

This is because for a compact manifold, there is no boundary and hence by Gauss' theorem the integral $\int_{\mathcal{M}} \sqrt{G} (D \cdot v) d^d$ over the the boundary of the target space will vanish. If we consider (5.90), this equation tells us that at a point p where R takes its minimum value on $\mathcal{M}$, the right-hand side of the equation is negative and the left-hand side is positive. Therefore, both $R_{\mu\nu}(p) = R(p) = 0$, but we also know that $R \geq 0$ on all of $\mathcal{M}$. Then using (5.91) we get $R = 0$ everywhere and this implies from (5.90) that $R_{\mu\nu} = 0$. This implies that at one-loop, a compact Riemannian manifold can be a target space for a sigma model as long as it is Ricci flat for the theory to be conformally invariant. But this is only for one-loop, we can see from the expansion of the Polyakov action (5.59) that if we compute higher-order corrections, there can be curvature terms like $R_{\mu\nu\sigma\rho}R^{\mu\nu\sigma\rho}$ and the vanishing of the Ricci tensor does not constrain these higher-order curvature terms. This implies that there is no compact Riemannian manifold that can serve as the target space for the 2D sigma model that is conformally invariant to all orders in perturbation theory.

This is in contrast to the Zamolodchikov-Polchinski proof we explored in chapter 3 and 4 that in two-dimensional quantum field theories, scale invariance implies conformal invariance. But in the case of the two-dimensional nonlinear sigma model, we demonstrated that this does not necessarily hold. Although we can have scale invariance, which requires that the target space be quasi-Ricci flat, conformal invariance requires that the target space be Ricci flat at one loop.

Chapter 6

Conclusion

We illustrated a rigorous argument due to Zamalodchikov that in two-dimensions under the assumptions of Poincaré invariance, renormalizability, unitarity, and discreteness of the spectrum scale invariance always implies conformal invariance. We now outline the various physical applications of this result

1. Critical phenomena: The study of phase transition has been of utmost interest in statistical mechanics and various other areas of physics. The second order phase transition is a known example of critical phenomena. At the second order phase transition the various thermodynamic quantities obey relations of form

$$C \propto (T - T_c)^{-\alpha}. \quad (6.1)$$

This surprising fact led to the idea of universality. This basically means that all the theories which flow to a similar type of fixed point have same critical exponents. An important question is to understand this universality behaviour of the theories. These critical phenomena are described a renormalization group fixed point and one crucial observation is that at this point thermodynamic quantities show the scaling behaviour, this is the scaling hypothesis. Scaling hypothesis gives allows us to compute almost all of the critical exponents.

This is where the power of conformal invariance comes into play. If we assume that critical phenomena are classified by conformally invariant fixed point. Then classification of critical phenomena is equivalent to the classification of various conformal field theories. The conformal invariance then solely determines the critical exponents of the universality class. In two-dimensions a complete classification of critical phenomena described by conformal field theory is possible.

2. String theory: The worldsheet formalism of the first quantized string theory requires conformal invariance. We know that in path integral formulation of string theory we have to introduce ghosts fields, in order to get rid of these ghosts and to consistently compute amplitudes the worldsheet must be conformally invariant. Although in the chapter 5 we showed that the condition for scale invariance and conformal invariance can be different. This implies that there are target spaces which are not consistent as string background. This is because the worldsheet theory violates the assumptions.

Another important application is in holography (gauge/gravity duality), where a quantum theory of gravity in an anti-de Sitter (AdS) spacetime is mathematically equivalent to a conformal field theory without gravity that lives on the boundary.

3. Particle physics: Quantum field theory provides the mathematical framework for understanding particle physics. The best known example of conformal field theory realized in nature is the Maxwell's theory of electromagnetism. In an $SU(N)$ gauge

theory with fermions in the fundamental representation the β -function is

$$\beta(g) = -\frac{g^3}{16\pi^2} \left(\frac{11}{3}N - \frac{2}{3}n_f \right), \quad (6.2)$$

where n_f is the number of fermions. The negative sign implies that for sufficiently small number of fermions (n_f), the non-Abelian gauge theory is asymptotically free. This free theory is asymptotic to the Gaussian scale invariant fixed point in the UV limit. We have seen earlier that the Gaussian fixed point is conformally invariant. But it is not in general true that the scale invariant fixed point will be conformally invariant. An example would be to consider Einstein-Hilbert action, at linearized level is scale invariant but not conformally invariant.

Although these are only a few examples which demonstrate why an enhanced conformal invariance leads to some powerful insight about the system we are dealing with. We showed in chapter 3 that using conformal perturbation theory we can compute some interesting features about the renormalization group behaviour under certain simple assumptions.

In chapter 5 we demonstrated that there is no compact Riemannian manifold that can serve as the target space for $2D$ non-linear sigma model that is conformally invariant to all orders in perturbation theory. The argument for non-existence of such a solution relied on compactness in an essential way. At one-loop the argument due to Zamolodchikov still holds because any quasi-Ricci flat manifold must Ricci flat if the manifold is compact.

It will be interesting to explore how the proof will be modified in case of non-compact spaces. The improvement term in stress tensor in chapter 5 required the invertibility of the matrix of anomalous dimension which is true in case of compact manifold but does not necessarily hold for non-compact spaces. It is not immediately clear how the proof would go through, it is also not clear if the expectation values in equation (3.41) are well defined in case of non-compact and inhomogeneous manifold. If the functions are not well defined then the proof will fail.

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